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November 25, 2005

Chapter 8

Black-Scholes

8.1 Introduction

In this chapter we derive the *Black-Scholes formulas* for the price of a call option and the price of a put option as the limit of the option prices in an N -period binomial model as the number of steps N goes to infinity. We also derive the *Black-Scholes partial differential equation*, and we verify that the Black-Scholes formulas are solutions of the Black-Scholes partial differential equation. We discuss the “Greeks,” the partial derivatives of the function given by the Black-Scholes formulas.

To take the limit in an N -period binomial model, we need two major theorems from probability theory, the *Law of Large Numbers* and the *Central Limit Theorem*. We present these in the next two sections, and in Section 8.4, we use them to obtain the Black-Scholes formulas.

8.2 The Law of Large Numbers

Theorem 8.2.1 *Let $X_1, X_2, \dots$ be a sequence of independent random variables, all with the same distribution. Assume each random variable has expected value m and a finite variance. (All the random variables must have the same expected value and variance because they all have the same distribution.) Then the probability is one that*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N X_n = m. \quad (8.2.1)$$

For the limit of an N -period binomial model, we will use the Law of Large Numbers in the context of coin tossing. Suppose we toss a fair coin repeatedly and count the number of heads. Indeed, let

$$X_n = \begin{cases} +1 & \text{if the } n\text{-th coin toss results in } H, \\ -1 & \text{if the } n\text{-th coin toss results in } T. \end{cases} \quad (8.2.2)$$

These random variables X_n are independent and all have the same distribution. The expected value of each X_n is 1 and the variance of each X_n is also 1. We define

$$M_N = \sum_{n=1}^N X_n, \quad (8.2.3)$$

and call the process $M_n, M_1, M_2, \dots$ a *symmetric random walk*. The Strong Law of Large numbers says that the probability is one that

$$\lim_{N \rightarrow \infty} \frac{M_N}{N} = 0. \quad (8.2.4)$$

We also consider the random variables X_n^+ , which are given by

$$X_n^+ = \begin{cases} 1 & \text{if the } n\text{-th coin toss results in } H, \\ 0 & \text{if the } n\text{-th coin toss results in } T. \end{cases}$$

The random variables $X_1^+, X_2^+, \dots$ are independent and identically distributed. They all have expected value $\frac{1}{2}$, and their common variance, $\frac{1}{4}$, is finite. We define

$$H_N = \sum_{n=1}^N X_n^+, \quad (8.2.5)$$

which is the number of heads in the first N tosses. The Law of Large Numbers applied to the sequence $X_1^+, X_2^+, \dots$, implies that with probability one,

$$\lim_{N \rightarrow \infty} \frac{H_N}{N} = \frac{1}{2}. \quad (8.2.6)$$

The ratio of the number of heads to the number of tosses converges to $\frac{1}{2}$.

Finally, we consider the random variables $X_n^+ - X_n$, which are given by

$$X_n^+ - X_n = \begin{cases} 0 & \text{if the } n\text{-th coin toss results in } H, \\ 1 & \text{if the } n\text{-th coin toss results in } T. \end{cases}$$

The random variables $X_1^+ - X_1, X_2^+ - X_2, \dots$ are independent and identically distributed. They all have expected value $\frac{1}{2}$, and their common variance, $\frac{1}{4}$, is finite. We define

$$T_N = \sum_{n=1}^N (X_n^+ - X_n), \quad (8.2.7)$$

which is the number of tails in the first N tosses. The Law of Large Numbers applied to the sequence $X_1^+ - X_1, X_2^+ - X_2, \dots$ implies that with probability one,

$$\lim_{N \rightarrow \infty} \frac{T_N}{N} = \frac{1}{2}. \quad (8.2.8)$$

The ratio of the number of tails to the number of tosses converges to $\frac{1}{2}$.

Finally, note that $X_n^+ + (X_n^+ - X_n)$ is equal to 1, regardless of whether the n -th toss results in H or T . Therefore,

$$H_N + T_N = \sum_{n=1}^N [X_n^+ + (X_n^+ - X_n)] = N. \quad (8.2.9)$$

This just says that in N tosses, the number of heads plus the number of tails is equal to the number of tosses. On the other hand,

$$H_N - T_N = \sum_{n=1}^N [X_n^+ - (X_n^+ - X_n)] = \sum_{n=1}^N X_n = M_N. \quad (8.2.10)$$

We said we were tossing a “fair” coin, by which we mean that the probability $\tilde{p}$ of a H on each toss is $\frac{1}{2}$ and hence the probability $\tilde{q}$ of a T is also $\frac{1}{2}$. We use the notation of risk-neutral probabilities $\tilde{p}$, $\tilde{q}$, $\tilde{\mathbb{P}}$ and $\tilde{\mathbb{E}}$ here because in the derivation of the Black-Scholes formulas in this chapter, we shall use the Law of Large Numbers under the risk-neutral probability measure.

Equation (8.2.6) says we should expect about half the coin tosses to result in H . This does not mean that if H gets ahead in the first several coin tosses, there is a need for T to catch up. For example, if we toss the coin 10 times and get a H on every toss, there is no need to then get more tails than heads on subsequent tosses. In particular, it is not true that the probability of a T on the eleventh toss is more than $\frac{1}{2}$. The tosses are independent, and the outcome of the first ten tosses has no bearing on the eleventh toss. The probability of a T on the eleventh toss is still $\frac{1}{2}$. Consider the case that there are 10 heads in the first ten tosses. If from that point on

there are as many tails as heads, so that T never catches up, then after 100 tosses there are 55 heads and $\frac{H_{100}}{100} = 0.55$, after 1000 tosses there are 505 heads and $\frac{H_{1000}}{1000} = 0.505$, and after 10,000 tosses there are 5005 heads and $\frac{H_{10,000}}{10,000} = 0.5005$. The ratio $\frac{H_N}{N}$ is converging to $\frac{1}{2}$, even though the number of heads is always ahead of the number of tails by the same amount.

In fact, if we toss the coin N times, it is normal to have the number of heads differ from $\frac{N}{2}$ by about $\sqrt{N}$. In 100 tosses, we should not be surprised if the number of heads differs from 50 by about $\sqrt{100} = 10$. In 10,000 tosses, we should not be surprised if the number of heads differs from 5,000 by about $\sqrt{10,000} = 100$. Note that if we have $50 + 10 = 60$ heads in the first 100 tosses, then $\frac{H_{100}}{100} = 0.60$. If we have $5000 + 100 = 5100$ heads in the first 10,000 tosses, then $\frac{H_{10,000}}{10,000} = 0.51$. The ratio $\frac{H_N}{N}$ is converging to $\frac{1}{2}$, even though H_N is getting farther from $\frac{N}{2}$ as N gets bigger. The Law of Large Numbers does not guarantee that the number of heads in the first N tosses is close to $\frac{N}{2}$, and in fact, these two quantities typically get farther apart rather than closer as $N \rightarrow \infty$.

8.3 The Central Limit Theorem

In the discussion of fair coin tossing in Section 8.2, we saw that

$$\lim_{N \rightarrow \infty} \frac{H_N}{N} = \frac{1}{2}, \quad (8.2.6)$$

but that this does not guarantee that H_N and $\frac{N}{2}$ are close. However, there are some conclusions we can draw from (8.2.6). One of them is that if we divide by N to a power larger than 1 in (8.2.6), we will get a limit of zero. For example,

$$\lim_{N \rightarrow \infty} \frac{H_N}{N\sqrt{N}} = \lim_{N \rightarrow \infty} \frac{H_N}{N} \cdot \lim_{N \rightarrow \infty} \frac{1}{\sqrt{N}} = \frac{1}{2} \cdot 0 = 0. \quad (8.3.1)$$

If we divide by N to a power between 0 and 1, we will get ∞ . For example,

$$\lim_{N \rightarrow \infty} \frac{H_N}{\sqrt{N}} = \lim_{N \rightarrow \infty} \frac{H_N}{N} \cdot \lim_{N \rightarrow \infty} \sqrt{N} = \frac{1}{2} \cdot \lim_{N \rightarrow \infty} \sqrt{N} = \infty. \quad (8.3.2)$$

If we begin instead with (8.2.4),

$$\lim_{N \rightarrow \infty} \frac{M_N}{N} = 0, \quad (8.2.4)$$

and replace N in the denominator by different powers of N , the situation is more complicated. Analogous to (8.3.1), we have

$$\lim_{N \rightarrow \infty} \frac{M_N}{N\sqrt{N}} = \lim_{N \rightarrow \infty} \frac{M_N}{N} \cdot \lim_{N \rightarrow \infty} \frac{1}{\sqrt{N}} = 0 \cdot 0 = 0. \quad (8.3.3)$$

However,

$$\lim_{N \rightarrow \infty} \frac{M_N}{\sqrt{N}} = \lim_{N \rightarrow \infty} \frac{M_N}{N} \cdot \lim_{N \rightarrow \infty} \sqrt{N}, \quad (8.3.4)$$

and this leads to the indeterminate form $0 \cdot \infty$. In such a situation, the limit could be anything, and could even fail to exist. In fact, if we toss a coin repeatedly and write down the resulting sequence

$$M_1, \frac{M_2}{\sqrt{2}}, \frac{M_3}{\sqrt{3}}, \frac{M_4}{\sqrt{4}}, \dots, \quad (8.3.5)$$

this sequence of numbers will never settle down and have a limit.

Despite that fact that the sequence in (8.3.5) does not have limit, we can say something about what happens to $\frac{M_N}{\sqrt{N}}$ as N gets large. We can plot the histogram of the distribution of the random variable $\frac{M_N}{\sqrt{N}}$, and we discover that as N gets large, this histogram takes a particular shape, namely, the “bell-shaped curve.”

We work out the details for the case $N = 25$. In this case,

$$\frac{M_{25}}{\sqrt{25}} = \frac{1}{5}M_{25}.$$

In 25 tosses, the number of heads that can occur is any integer between 0 and 25. If $H_{25} = 0$, then $T_{25} = 25$ and $M_{25} = H_{25} - T_{25} = -25$, so $\frac{1}{5}M_{25} = -5$. This is the smallest possible value for $\frac{1}{5}M_{25}$. If $H_{25} = 1$, then $T_{25} = 24$, $M_{25} = -23$, and $\frac{1}{5}M_{25} = -\frac{23}{5} = -4.6$. With each increase in H_{25} of one head, there is a decrease in T_{25} of 1 and hence an increase in M_{25} of 2 and an increase in $\frac{1}{5}M_{25}$ of $\frac{2}{5} = 0.4$. At the upper extreme, if $H_{25} = 25$, then $T_{25} = 0$, $M_{25} = 25$ and $\frac{1}{5}M_{25} = +5$. This is the largest possible value for $\frac{1}{5}M_{25}$. The set of possible values for $\frac{1}{5}M_{25}$ is thus

$$x_0 = -5, x_1 = -4.6, x_2 = -4.2, \dots, x_{12} = -0.2, x_{13} = 0.2, \dots, x_{25} = 5.$$

The probabilities that $\frac{1}{5}M_{25}$ takes these values are given by the formula

$$\tilde{\mathbb{P}} \left\{ \frac{1}{5}M_{25} = x_k \right\} = \frac{25!}{k!(25-k)!} \cdot \frac{1}{2^{25}}, \quad k = 0, 1, \dots, 25. \quad (8.3.6)$$

We record these probabilities in Table 8.3.1 below.

k	x_k	$\tilde{\mathbb{P}}\{\frac{1}{5}M_{25} = x_k\}$	h_k	$\varphi(x_k)$
0	-5.0	2.980×10^{-8}	7.451×10^{-8}	1.487×10^{-6}
1	-4.6	7.451×10^{-7}	1.863×10^{-6}	1.014×10^{-5}
2	-4.2	8.941×10^{-6}	2.235×10^{-5}	5.894×10^{-5}
3	-3.8	6.855×10^{-5}	1.714×10^{-4}	2.919×10^{-4}
4	-3.4	3.770×10^{-4}	9.425×10^{-4}	1.232×10^{-3}
5	-3.0	0.001583	0.003958	0.0044
6	-2.6	0.005278	0.01319	0.0136
7	-2.2	0.01433	0.03581	0.0355
8	-1.8	0.03223	0.08058	0.0790
9	-1.4	0.06089	0.1522	0.1497
10	-1.0	0.09742	0.2435	0.2420
11	-0.6	0.1328	0.3321	0.3332
12	-0.2	0.1550	0.3875	0.3910
13	0.2	0.1550	0.3875	0.3910
14	0.6	0.1328	0.3321	0.3332
15	1.0	0.09742	0.2435	0.2420
16	1.4	0.06089	0.1522	0.1497
17	1.8	0.03223	0.08058	0.0790
18	2.2	0.01433	0.03581	0.0355
19	2.6	0.005278	0.01319	0.0136
20	3.0	0.001583	0.003958	0.0044
21	3.4	3.770×10^{-4}	9.425×10^{-4}	1.232×10^{-3}
22	3.8	6.855×10^{-5}	1.714×10^{-4}	2.919×10^{-4}
23	4.2	8.941×10^{-6}	2.235×10^{-5}	5.894×10^{-5}
24	4.6	7.451×10^{-7}	1.863×10^{-6}	1.014×10^{-5}
25	5.0	2.980×10^{-8}	7.451×10^{-8}	1.487×10^{-6}

Table 8.3.1

We can use this table to construct the histogram in Figure 8.3.2. Above each of the points x_k we construct a bar. The width of each bar is 0.4. For example, the bar constructed above $x_{13} = 0.2$ has its left side at 0.0 and its right side at 0.4. The adjacent bar, the one constructed above $x_{14} = 0.6$, shares the side at 0.4 with the bar constructed above $x_{13} = 0.2$ and has its right side at 0.8. The width of the bar constructed above $x_{14} = 0.6$ is $0.8 - 0.4 = 0.4$.

We construct the bars in the histogram so that the area in the bar

above each x_k is the $\tilde{\mathbb{P}}\{\frac{1}{5}M_{25} = x_k\}$. This means that the height of the bar above x_k is

$$h_k = \frac{1}{0.4} \tilde{\mathbb{P}}\left\{\frac{1}{5}M_{25} = x_k\right\}. \quad (8.3.7)$$

In Table 8.3.1, we record the values of h_k as well as the probabilities $\tilde{\mathbb{P}}\{\frac{1}{5}M_{25} = x_k\}$ given by (8.3.6).

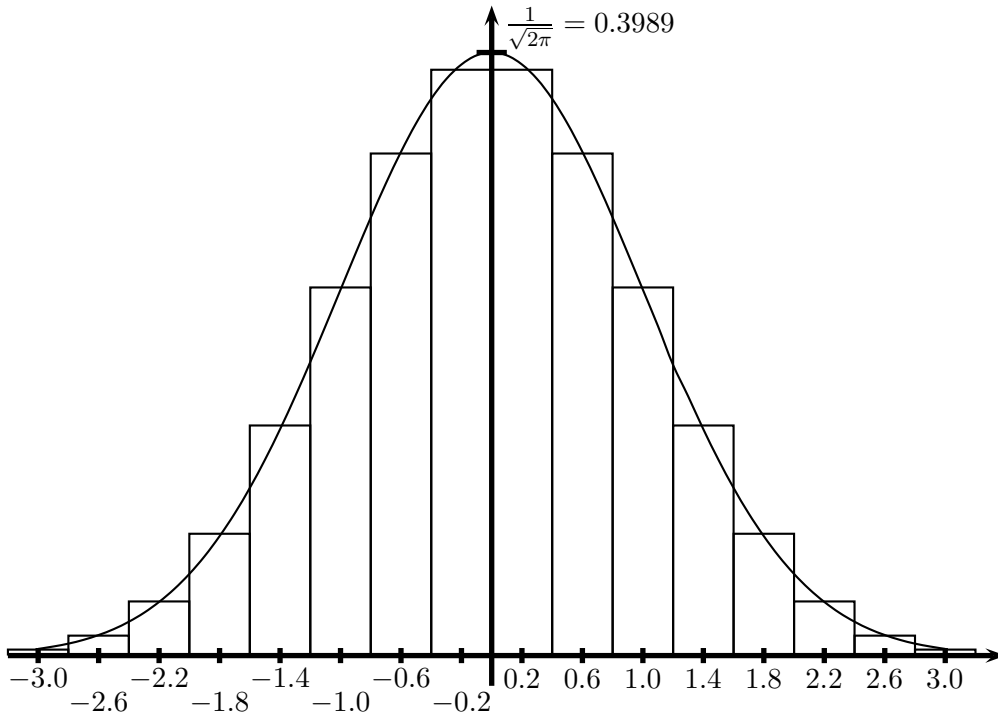


Figure 8.3.2: Histogram for $\frac{1}{5}M_{25}$ with normal curve $y = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$.

The *standard normal density* is

$$\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}. \quad (8.3.8)$$

The values of this function are reported in the last column of Table 8.3.1, and the graph of this function is superimposed on the histogram in Figure 8.3.2. We see that this function is a good approximation to the heights of the bars in the histogram. In particular

$$h_k \approx \varphi(x_k), \quad k = 0, 1, \dots, 25, \quad (8.3.9)$$

where $\approx$ means “is approximately equal to.”

Suppose that for some continuous function $f(x)$, we want to evaluate

$$\tilde{\mathbb{E}} \left[f \left(\frac{1}{5} M_{25} \right) \right].$$

This would require that we compute the probabilities of $\tilde{\mathbb{P}}\{\frac{1}{5} M_{25} = x_k\}$, as we have done, and then evaluate the sum

$$\tilde{\mathbb{E}} \left[f \left(\frac{1}{5} M_{25} \right) \right] = \sum_{k=0}^{25} f(x_k) \tilde{\mathbb{P}} \left\{ \frac{1}{5} M_{25} = k \right\} = \sum_{k=0}^{25} f(x_k) (h_k \times 0.4). \quad (8.3.10)$$

This is already a long computation when the number of tosses is 25, as in (8.3.10), and it becomes extremely time consuming when the number of tosses is larger, say $N = 100$ or $N = 1000$. Fortunately, we can avoid this computation because we can get a good approximation to the expected value in (8.3.10) by using (8.3.9) to replace h_k in (8.3.10) and by $\varphi(x_k)$:

$$\tilde{\mathbb{E}} \left[f \left(\frac{1}{5} M_{25} \right) \right] \approx \sum_{k=0}^{25} f(x_k) \varphi(x_k) \times 0.4. \quad (8.3.11)$$

The right-hand side of (8.3.11) is an approximating sum for a Riemann integral. In fact, if we wanted to approximate the Riemann integral

$$\int_{-\infty}^{\infty} f(x) \varphi(x) dx, \quad (8.3.12)$$

as a sum, we would choose some points on the real line, say $x_0 = -5.0$, $x_1 = -4.6, \dots, x_{25} = 5.0$, and above each of these points we would build a rectangle. Since the distance between the points is 0.4, we could build the rectangles to be centered at these points and each with width 0.4. Since $f(x)\varphi(x)$ is the function we want to integrate, we would make the height of the rectangle at x_k equal to $f(x_k)\varphi(x_k)$. This would result in the sum on the right-hand side of (8.3.11), which is therefore an approximation to the Riemann integral

$$\int_{-5}^5 f(x) \varphi(x) dx. \quad (8.3.13)$$

But when $|x| > 5$, $\varphi(x)$ is very small (see the last column in Table 8.3.1), and so, provided $f(x)$ does not grow too rapidly as $x \rightarrow \infty$ or $x \rightarrow -\infty$, the difference between the integral in (8.3.13) and the integral in (8.3.12)

is small. In general, we will use this approximation only for functions $f(x)$ that satisfy

$$|f(x)| \leq C \text{ for all } x \in \mathbb{R} \quad (8.3.14)$$

for some constant C . Such a function is said to be *bounded*, and for such a function, the difference between (8.3.13) and (8.3.12) can be ignored when the number of coin tosses is large. For a bounded function, we have the approximation

$$\tilde{\mathbb{E}} \left[f \left(\frac{1}{5} M_{25} \right) \right] \approx \int_{-\infty}^{\infty} f(x) \varphi(x) dx. \quad (8.3.15)$$

Saying that two things are approximately equal, as we just did in (8.3.15), is not a precise mathematical statement. We make precise the idea we are trying to capture in (8.3.15) using limits. The precise statement for the situation we have been discussing is the Central Limit Theorem, which we now state.

Theorem 8.3.1 (Central Limit) *Let $X_1, X_2, \dots$ be a sequence of independent, identically distributed random variables under a probability measure $\tilde{\mathbb{P}}$. Assume that $\tilde{\mathbb{E}}X_n = 0$ and $\text{Var}(X_n) = 1$. (The expected value and the variance is the same for each X_n because all these random variables have the same distribution.) Let $f(x)$ be a bounded function defined on the real line that is continuous except possibly at finitely many points. Then*

$$\lim_{N \rightarrow \infty} \tilde{\mathbb{E}} \left[f \left(\frac{1}{\sqrt{N}} \sum_{n=1}^N X_n \right) \right] = \int_{-\infty}^{\infty} f(x) \varphi(x) dx. \quad (8.3.16)$$

It can happen that the random variables we wish to study are of the form

$$\frac{\gamma}{\sqrt{N}} \sum_{n=1}^N X_n + Y_N,$$

where the sequence $X_1, X_2, \dots$ is as in the Central Limit Theorem, γ is a real number, and $Y_1, Y_2, \dots$ is a sequence of random variables converging to a real number y . For such a case, we have the following generalization of the Central Limit Theorem.

Theorem 8.3.2 (Generalized Central Limit) *Let $X_1, X_2, \dots$ be a sequence of independent, identically distributed random variables under a probability measure $\tilde{\mathbb{P}}$. Assume that $\tilde{\mathbb{E}}X_n = 0$ and $\text{Var}(X_n) = 1$. Let γ be a real*

number and let $Y_1, Y_2, \dots$ be a sequence of random variables such that, with probability 1,

$$\lim_{N \rightarrow \infty} Y_N = y,$$

where y is a real number. Let $f(x)$ be a bounded function defined on the real line that is continuous except possibly at finitely many points. then

$$\lim_{N \rightarrow \infty} \widetilde{\mathbb{E}} \left[f \left(\frac{\gamma}{\sqrt{N}} \sum_{n=1}^N X_n + Y_N \right) \right] = \int_{-\infty}^{\infty} f(\gamma x + y) \varphi(x) dx. \quad (8.3.17)$$

We close this section with a few observations about the standard normal density $\varphi(x)$. This function is positive for every $x \in \mathbb{R}$, and φ integrates to one:

$$\int_{-\infty}^{\infty} \varphi(x) dx = 1. \quad (8.3.18)$$

We shall use (8.3.18) without deriving it. (To derive (8.3.18), write

$$\begin{aligned} \left(\int_{-\infty}^{\infty} \varphi(x) dx \right)^2 &= \int_{-\infty}^{\infty} \varphi(x) dx \cdot \int_{-\infty}^{\infty} \varphi(y) dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy, \end{aligned}$$

and change to polar coordinates to compute the right-hand side, which turns out to be 1.)

The *cumulative standard normal distribution* is

$$N(x) = \int_{-\infty}^x \varphi(y) dy. \quad (8.3.19)$$

For $x \in \mathbb{R}$, $N(x)$ is the area under the graph of φ to the left of the point x . Because φ is symmetric, (i.e., $\varphi(y) = \varphi(-y)$ for all $y \in \mathbb{R}$), the area under the graph of φ to the left of x , which is $N(x)$, is the same as the area under the graph of φ to the right of $-x$. But the total area under the graph of φ is 1 and the area under the graph to the left of $-x$ is $N(-x)$, so the area under the graph to the right of $-x$ is $1 - N(-x)$. In other words,

$$N(x) = 1 - N(-x) \text{ for all } x \in \mathbb{R}. \quad (8.3.20)$$

From the definition of $N(x)$, we see that

$$N'(x) = \varphi(x). \quad (8.3.21)$$

In particular, $N'(x)$ is strictly positive, so N is strictly increasing. From the definition of $N(x)$, it is apparent that

$$\lim_{x \rightarrow -\infty} N(x) = 0. \quad (8.3.22)$$

From (8.3.18), we have

$$\lim_{x \rightarrow \infty} N(x) = \int_{-\infty}^{\infty} \varphi(y) dy = 1. \quad (8.3.23)$$

8.4 The Black-Scholes Formulas

In this section we develop the Black-Scholes formula for European puts and calls.

8.4.1 Scaling of interest rate and volatility

We consider a stock with initial price per share S_0 . On this stock we have a put option expiring at a positive time τ (measured in years) and having a positive strike price K . We divide the time between 0 and τ into N steps, so that each step corresponds to a period of time of length $\frac{\tau}{N}$.¹

We want to build a binomial model which has N steps between time zero and time τ . To simplify the computations, we will design this model so that the risk-neutral probabilities are $\tilde{p} = \tilde{q} = \frac{1}{2}$. If the up factor per period is u and the down factor is d , where $0 < d < 1 + r < u$, then in order to have

$$\tilde{p} = \frac{1 + r - d}{u - d} = \frac{1}{2}, \quad \tilde{q} = \frac{u - 1 - r}{u - d} = \frac{1}{2},$$

we must have

$$2(1 + r - d) = u - d = 2(u - 1 - r),$$

and so $u - (1 + r) = (1 + r) - d$. We call this common value σ , which is positive. In other words,

$$\sigma = u - (1 + r) = (1 + r) - d, \quad (8.4.1)$$

¹A stock price can change only when it is possible to trade it, which generally means only when the exchange on which it is traded is open. In practice, one needs to account for this fact when dividing the time interval between zero and τ into steps. For example, suppose $\tau = \frac{1}{4}$ years, so we have a “three-month option.” In three months there are approximately 63 trading days. Thus, If we take $N = 63$, we are dividing time into steps of one trading day each and are excluding non-trading days from consideration. In this section we ignore these so-called *day count* issues.

or equivalently,

$$u = 1 + r + \sigma, \quad d = 1 + r - \sigma. \quad (8.4.2)$$

The risk-neutral expected return between time n and time $n + 1$ is defined to be

$$\begin{aligned} \tilde{\mathbb{E}} \left[\frac{S_{n+1} - S_n}{S_n} \right] &= \tilde{p} \frac{uS_n - S_n}{S_n} + \tilde{q} \frac{dS_n - S_n}{S_n} \\ &= \frac{1}{2}(u - 1) + \frac{1}{2}(d - 1) \\ &= \frac{1}{2}(r + \sigma) + \frac{1}{2}(r - \sigma) \\ &= r, \end{aligned} \quad (8.4.3)$$

which is what the expected return must be under the risk-neutral measure. To determine the risk-neutral variance of the return between time n and time $n + 1$, we first compute

$$\begin{aligned} \tilde{\mathbb{E}} \left[\left(\frac{S_{n+1} - S_n}{S_n} \right)^2 \right] &= \tilde{p} \left(\frac{uS_n - S_n}{S_n} \right)^2 + \tilde{q} \left(\frac{dS_n - S_n}{S_n} \right)^2 \\ &= \frac{1}{2}(u - 1)^2 + \frac{1}{2}(d - 1)^2 \\ &= \frac{1}{2}(r + \sigma)^2 + \frac{1}{2}(r - \sigma)^2 \\ &= r^2 + \sigma^2. \end{aligned} \quad (8.4.4)$$

The risk-neutral variance of the return is

$$\begin{aligned} \text{Var} \left[\frac{S_{n+1} - S_n}{S_n} \right] &= \tilde{\mathbb{E}} \left[\left(\frac{S_{n+1} - S_n}{S_n} \right)^2 \right] - \left(\tilde{\mathbb{E}} \left[\frac{S_{n+1} - S_n}{S_n} \right] \right)^2 \\ &= (r^2 + \sigma^2) - r^2 \\ &= \sigma^2. \end{aligned} \quad (8.4.5)$$

Having thus computed the one-period expected return and variance of return, we next compute the expected return and variance of return for N periods. To do this, we need the formulas

$$\mathbb{E} \left[\frac{S_{n+1}}{S_n} \right] = \tilde{p}u + \tilde{q}d = \frac{1}{2}(1 + r + \sigma) + \frac{1}{2}(1 + r - \sigma) = 1 + r \quad (8.4.6)$$

and

$$\tilde{\mathbb{E}} \left[\frac{S_{n+1}^2}{S_n^2} \right] = \tilde{p}u^2 + \tilde{q}d^2 = \frac{1}{2}(1 + r + \sigma)^2 + \frac{1}{2}(1 + r - \sigma)^2 = (1 + r)^2 + \sigma^2. \quad (8.4.7)$$

The actual return over the N periods beginning at time zero is $\frac{S_N - S_0}{S_0}$. Using the independence of the random variables $\frac{S_1}{S_0}, \frac{S_2}{S_1}, \dots, \frac{S_{N-1}}{S_{N-2}}, \frac{S_N}{S_{N-1}}$ and using (8.4.6), we compute

$$\begin{aligned} \tilde{\mathbb{E}} \left[\frac{S_N}{S_0} \right] &= \tilde{\mathbb{E}} \left[\frac{S_1}{S_0} \cdot \frac{S_2}{S_1} \cdot \dots \cdot \frac{S_{N-1}}{S_{N-2}} \cdot \frac{S_N}{S_{N-1}} \right] \\ &= \tilde{\mathbb{E}} \left[\frac{S_1}{S_0} \right] \cdot \tilde{\mathbb{E}} \left[\frac{S_2}{S_1} \right] \cdot \dots \cdot \tilde{\mathbb{E}} \left[\frac{S_{N-1}}{S_{N-2}} \right] \cdot \tilde{\mathbb{E}} \left[\frac{S_N}{S_{N-1}} \right] \\ &= (1+r)^N. \end{aligned} \tag{8.4.8}$$

The expected return over the N periods beginning at time zero is

$$\tilde{\mathbb{E}} \left[\frac{S_N - S_0}{S_0} \right] = \tilde{\mathbb{E}} \left[\frac{S_N}{S_0} \right] - 1 = (1+r)^N - 1. \tag{8.4.9}$$

To compute the variance of the expected return, we first use (8.4.8), independence of $\frac{S_1^2}{S_0^2}, \frac{S_2^2}{S_1^2}, \dots, \frac{S_{N-1}^2}{S_{N-2}^2}, \frac{S_N^2}{S_{N-1}^2}$, and (8.4.7) to compute

$$\begin{aligned} &\tilde{\mathbb{E}} \left[\left(\frac{S_N - S_0}{S_0} \right)^2 \right] \\ &= \tilde{\mathbb{E}} \left[\frac{S_N^2 - 2S_N S_0 + S_0^2}{S_0^2} \right] \\ &= \tilde{\mathbb{E}} \left[\frac{S_N^2}{S_0^2} \right] - 2\tilde{\mathbb{E}} \left[\frac{S_N}{S_0} \right] + 1 \\ &= \tilde{\mathbb{E}} \left[\frac{S_1^2}{S_0^2} \cdot \frac{S_2^2}{S_1^2} \cdot \dots \cdot \frac{S_{N-1}^2}{S_{N-2}^2} \cdot \frac{S_N^2}{S_{N-1}^2} \right] - 2(1+r)^N + 1 \\ &= \tilde{\mathbb{E}} \left[\frac{S_1^2}{S_0^2} \right] \cdot \tilde{\mathbb{E}} \left[\frac{S_2^2}{S_1^2} \right] \cdot \dots \cdot \tilde{\mathbb{E}} \left[\frac{S_{N-1}^2}{S_{N-2}^2} \right] \cdot \tilde{\mathbb{E}} \left[\frac{S_N^2}{S_{N-1}^2} \right] - 2(1+r)^N + 1 \\ &= ((1+r)^2 + \sigma^2)^N - 2(1+r)^N + 1. \end{aligned} \tag{8.4.10}$$

The variance of the N -period return is

$$\begin{aligned}
& \text{Var} \left[\frac{S_N - S_0}{S_0} \right] \\
&= \widetilde{\mathbb{E}} \left[\left(\frac{S_N - S_0}{S_0} \right)^2 \right] - \left(\widetilde{\mathbb{E}} \left[\frac{S_N - S_0}{S_0} \right] \right)^2 \\
&= ((1+r)^2 + \sigma^2)^N - 2(1+r)^N + 1 - ((1+r)^N - 1)^2 \\
&= ((1+r)^2 + \sigma^2)^N - 2(1+r)^N + 1 - ((1+r)^{2N} - 2(1+r)^N + 1) \\
&= ((1+r)^2 + \sigma^2)^N - (1+r)^{2N}. \tag{8.4.11}
\end{aligned}$$

We divide the time interval $[0, \tau]$ into N steps, each of which will have length $\frac{\tau}{N}$, and we ultimately let $N \rightarrow \infty$. When we do that, the average growth rate of the stock will be the continuously compounding rate. We would like the interest rate for this continuously compounding to be r per year. In particular, we want to have $\widetilde{\mathbb{E}}[S_N] = e^{r\tau} S_0$, or equivalently, we want to have the N -period risk-neutral expected return on the stock to be

$$\widetilde{\mathbb{E}} \left[\frac{S_N - S_0}{S_0} \right] = e^{r\tau} - 1. \tag{8.4.12}$$

But in order to have interest rate r per year, we cannot also have interest rate r per period, which is a fraction of year. Indeed, comparing (8.4.12) to (8.4.9), we see that the interest accrued over each period should be computed using $\frac{r\tau}{N}$ rather than r . This makes sense because when r is the annual interest rate, the interest accrued on \$1 over a time period of length $\frac{\tau}{N}$ should be $\frac{r\tau}{N}$. If we substitute $\frac{r\tau}{N}$ for r in (8.4.9) and then let $N \rightarrow \infty$, we obtain the desired expected return,

$$\lim_{N \rightarrow \infty} \widetilde{\mathbb{E}} \left[\frac{S_N - S_0}{S_0} \right] = \lim_{N \rightarrow \infty} \left(1 + \frac{r\tau}{N} \right)^N - 1 = e^{r\tau} - 1, \tag{8.4.13}$$

where the limit is justified by substituting $a = r\tau$ and $b = 0$ into Lemma 8.4.1, which follows.

Lemma 8.4.1 *Let a and b be real numbers. Then*

$$\lim_{N \rightarrow \infty} \left(1 + \frac{a}{N} + \frac{b}{N^2} \right)^N = e^a. \tag{8.4.14}$$

PROOF: We make the change of variable $x = \frac{1}{N}$ in the following calculation:

$$\begin{aligned} \lim_{N \rightarrow \infty} \ln \left(1 + \frac{a}{N} + \frac{b}{N^2} \right)^N &= \lim_{x \rightarrow 0} \ln (1 + ax + bx^2)^{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \frac{\ln(1 + ax + bx^2)}{x}. \end{aligned}$$

This leads to the indeterminate form $\frac{0}{0}$, and so we use L'Hopital's rule to compute

$$\lim_{x \rightarrow 0} \frac{\ln(1 + ax + bx^2)}{x} = \lim_{x \rightarrow 0} \frac{a + 2bx}{1 + ax + bx^2} = a.$$

We have computed the logarithm of the limit in (8.4.14) and gotten a , so the limit in (8.4.14) is e^a .

We also want to have a limiting variance for the stock return as $N \rightarrow \infty$. The variance is given by (8.4.11). We have already seen that we should replace r by $\frac{r\tau}{N}$. To guarantee that there is a meaningful limit as $N \rightarrow \infty$ in (8.4.11), we similarly replace σ^2 by $\frac{\sigma^2\tau}{N}$, or equivalently, replace σ by $\frac{\sigma\sqrt{\tau}}{\sqrt{N}}$. With this substitution and again using Lemma 8.4.1, we obtain the limiting variance of return

$$\begin{aligned} &\lim_{N \rightarrow \infty} \text{Var} \left[\frac{S_N - S_0}{S_0} \right] \\ &= \lim_{N \rightarrow \infty} \left(\left(1 + \frac{r\tau}{N} \right)^2 + \frac{\sigma^2\tau}{N} \right)^N - \lim_{N \rightarrow \infty} \left(1 + \frac{2r\tau}{2N} \right)^{2N} \\ &= \lim_{N \rightarrow \infty} \left(1 + \frac{(2r + \sigma^2)\tau}{N} + \frac{r^2\tau^2}{N^2} \right)^N - e^{2r\tau} \\ &= e^{(2r + \sigma^2)\tau} - e^{2r\tau}. \end{aligned} \tag{8.4.15}$$

Note that if we had instead replaced σ by $\frac{\sigma\tau}{N}$, then the limiting variance of $\frac{S_N - S_0}{S_0}$ would be zero, so that there would be no randomness left as $N \rightarrow \infty$.

For that reason, we replace σ by $\frac{\sigma\sqrt{\tau}}{\sqrt{N}}$ rather than by $\frac{\sigma\tau}{N}$.

We now return to the formula (8.4.2) for the up and down factors in the N period model. Let τ (measured in years) be the time until expiration of the option we are considering. When we divide the time interval $[0, \tau]$ into N periods, we take these to be

$$u = 1 + \frac{r\tau}{N} + \frac{\sigma\sqrt{\tau}}{\sqrt{N}}, \quad d = 1 + \frac{r\tau}{N} - \frac{\sigma\sqrt{\tau}}{\sqrt{N}}. \tag{8.4.16}$$

With these choices of u and d , the risk-neutral probabilities are still $\tilde{p} = \tilde{q} = \frac{1}{2}$.

The parameter σ in (8.4.16) is called the *volatility* of the stock. This parameter describes how much the stock price moves over time and is thus a measure of the risk associated with investing in the stock.

When one builds a binomial model for a stock price on an interval of time from 0 to τ , dividing $[0, \tau]$ into N steps, one first estimates the volatility from price data and then takes the up and down factors to be either those given by (8.4.16) or by

$$u = e^{\sigma\sqrt{\tau}/\sqrt{N}}, \quad d = e^{-\sigma\sqrt{\tau}/\sqrt{N}}. \quad (8.4.17)$$

The formulas (8.4.16) and (8.4.17) are quite close, as we now show.

Recall from Taylor's Theorem that if a function $f(x)$ has continuous first and second derivatives, then

$$f(x) = f(0) + f'(0)x + \frac{1}{2}f''(\xi)x^2, \quad (8.4.18)$$

where ξ is a point between 0 and x . So long as we restrict attention to $x \in [-1, 1]$, the term $f''(\xi)$ is bounded by a constant, and we may rewrite (8.4.18) as

$$f(x) = f(0) + f'(0)x + O(x^2), \quad (8.4.19)$$

where we use the notation $O(x^2)$ to denote any term whose absolute value is less than or equal to a constant times x^2 so long as $x \in [-1, 1]$.

Applying Taylor's Theorem to $f(x) = e^x$, for which $f'(0) = 1$, we obtain from (8.4.19) that

$$e^x = 1 + x + O(x^2). \quad (8.4.20)$$

If $N > \sigma^2\tau$, then we can replace x by $\pm\sigma\sqrt{\tau}/\sqrt{N}$, which is in $[-1, 1]$, and (8.4.20) yields

$$e^{\pm\sigma\sqrt{\tau}/N} = 1 \pm \frac{\sigma\sqrt{\tau}}{\sqrt{N}} + O\left(\frac{1}{N}\right), \quad (8.4.21)$$

where we use the notation $O(\frac{1}{N})$ to denote any term whose absolute value is less than or equal to $\frac{1}{N}$ times a constant that does not depend on N . From (8.4.21) we see that the choice of u and d in (8.4.17) is close to the choice in (8.4.16), and in fact the difference is no larger than a constant times $\frac{1}{N}$. Because the choices in (8.4.16) and (8.4.17) are so close for large values of N , either choice will lead to the same Black-Scholes formulas; we make the choice (8.4.16) because it makes the derivation of the formulas simpler.

Remark 8.4.2 We discuss in more detail the $O(\cdot)$ notation. When we write $O(\frac{1}{N})$, we mean a term that depends on N and whose absolute value is less than or equal to a constant (that does not depend on N) times $\frac{1}{N}$, and this is valid for all $N = 1, 2, \dots$. An example of such a term is $\frac{e^{-N}}{N}$, because

$$\frac{e^{-N}}{N} \leq \frac{1}{N} \text{ for all } N = 1, 2, \dots$$

The term $\frac{e^N}{N}$ is not $O(\frac{1}{N})$ because we cannot find a constant not depending on N that bounds the numerator e^N . An example of a term that we would call $O(\frac{1}{N^2})$ is

$$\frac{-2N}{1+N^3}.$$

Indeed, for all $N = 1, 2, \dots$,

$$\left| \frac{-2N}{1+N^3} \right| = \frac{1}{N^2} \cdot \frac{2N}{\frac{1}{N^2} + N} \leq \frac{1}{N^2} \cdot \frac{2N}{N} = \frac{2}{N^2},$$

which is the constant 2 times $\frac{1}{N^2}$. The sum of any two $O(\frac{1}{N^2})$ terms is also an $O(\frac{1}{N^2})$ term. For example, the sum of $\frac{e^{-N}}{N^2}$ and $\frac{-2N}{1+N^3}$ can be bounded as follows:

$$\left| \frac{e^{-N}}{N^2} + \frac{-2}{1+N^3} \right| \leq \frac{e^{-N}}{N^2} + \frac{2N}{1+N^3} \leq \frac{1}{N} + \frac{2}{N^2} = \frac{3}{N^2}.$$

We sometimes write

$$O\left(\frac{1}{N^2}\right) + O\left(\frac{1}{N^2}\right) = O\left(\frac{1}{N^2}\right).$$

There is no need to write $2O(\frac{1}{N^2})$ on the right-hand side because $O(\frac{1}{N^2})$ means a term that is bounded by $\frac{1}{N^2}$ times *an arbitrary constant*, and we can incorporate the 2 into this constant.

A term that is $O(\frac{1}{N^2})$ has limit zero as $N \rightarrow \infty$. In fact, because such a term has an N^2 in the denominator, even if we multiply such a term by N , we would still have an N in the denominator and hence would still have a term with limit zero as $N \rightarrow \infty$. We write this multiplication as

$$N \cdot O\left(\frac{1}{N^2}\right) = O\left(\frac{1}{N}\right).$$

For example, $N \cdot \frac{e^{-N}}{N^2} = \frac{e^{-N}}{N} = O(\frac{1}{N})$.

The term $\frac{e^{-N}}{N^2}$ is $O(\frac{1}{N^2})$, but it is also $O(\frac{1}{N})$. This is because $\frac{e^{-N}}{N^2}$ is less than or equal to the constant 1 times $\frac{1}{N}$, i.e.,

$$\frac{e^{-N}}{N^2} \leq \frac{1}{N^2} \leq \frac{1}{N} \text{ for all } N = 1, 2, \dots$$

In general, any term that is $O(\frac{1}{N^2})$ is also $O(\frac{1}{N})$.

The sum of a term that is $O(\frac{1}{N})$ and a term that is $O(\frac{1}{N^2})$ is a term that is $O(\frac{1}{N})$. For example, $\frac{-2N}{1+N^3}$ is $O(\frac{1}{N^2})$ and $\frac{e^{-N}}{N}$ is $O(\frac{1}{N})$. The absolute value of the sum can be bounded as follows:

$$\left| \frac{-2N}{1+N^3} + \frac{e^{-N}}{N} \right| \leq \frac{2N}{1+N^3} + \frac{e^{-N}}{N} \leq \frac{2}{N^2} + \frac{1}{N^2} \leq \frac{3}{N} \text{ for all } N = 1, 2, \dots$$

We can also consider terms of the form $O(\frac{1}{\sqrt{N}})$ and $O(\frac{1}{N\sqrt{N}})$. An example of an $O(\frac{1}{\sqrt{N}})$ term is $u - 1$, where u is given by (8.4.16). In fact,

$$u - 1 = \frac{r\tau}{N} + \frac{\sigma\sqrt{\tau}}{\sqrt{N}} \leq \frac{r\tau + \sigma\sqrt{\tau}}{\sqrt{N}}$$

because $\frac{r\tau}{N} \leq \frac{r\tau}{\sqrt{N}}$ for all $N = 1, 2, \dots$

8.4.2 Black-Scholes price of a put

Consider an N -period binomial model with up and down factors u and d given by (8.4.16) and with per-period interest rate $\frac{r\tau}{N}$. The risk-neutral probabilities are

$$\tilde{p} = \frac{1 + \frac{r\tau}{N} - d}{u - d} = \frac{\sigma\sqrt{\tau}/\sqrt{N}}{2\sigma\sqrt{\tau}/\sqrt{N}} = \frac{1}{2}, \quad \tilde{q} = 1 - \tilde{p} = \frac{1}{2}. \quad (8.4.22)$$

This model is the result of dividing τ years into N steps, and so S_N is the stock price at time τ . This stock price is

$$S_N = S_0 u^{H_N} d^{T_N}, \quad (8.4.23)$$

where H_N is the number of heads obtained in N coin tosses, and T_N is the number of tails. In this subsection we price a put, so we are interested in computing

$$P_0 = \frac{1}{(1 + \frac{r\tau}{N})^N} \tilde{\mathbb{E}}[(K - S_N)^+] = \frac{1}{(1 + \frac{r\tau}{N})^N} \tilde{\mathbb{E}}[(K - S_0 u^{H_N} d^{T_N})^+] \quad (8.4.24)$$

for some positive strike price K . In particular, we want to compute the limit in (8.4.24) as $N \rightarrow \infty$. We can use Lemma 8.4.1 to compute the limit of the discount term in (8.4.24). In fact, Lemma 8.4.1 implies

$$\lim_{N \rightarrow \infty} \frac{1}{(1 + \frac{r\tau}{N})^N} = e^{-r\tau}. \quad (8.4.25)$$

We use the Central Limit Theorem to compute the limit of the expected value in (8.4.24). In particular, we compute

$$\lim_{N \rightarrow \infty} \tilde{\mathbb{E}} \left[(K - S_0 u^{H_N} d^{T_N})^+ \right], \quad (8.4.26)$$

and this will result in the Black-Scholes formula for the price of a put. For large values of N , the put price in the N -period binomial model will be close to the price given by the Black-Scholes formula.

We first work out the Taylor series expansion for the function $f(x) = \ln(1+x)$. We need the first two derivatives and their values at zero, which are as follows:

$$\begin{aligned} f(x) &= \ln(1+x), & f(0) &= 0, \\ f'(x) &= \frac{1}{1+x}, & f'(0) &= 1, \\ f''(x) &= -\frac{1}{(1+x)^2}, & f''(0) &= -1. \end{aligned}$$

According to Taylor's Theorem,

$$\ln(1+x) = f(0) + x f'(0) + \frac{1}{2} x^2 f''(0) + O(x^3) = x - \frac{1}{2} x^2 + O(x^3). \quad (8.4.27)$$

Therefore,

$$\begin{aligned} \ln u &= \ln \left(1 + \frac{r\tau}{N} + \frac{\sigma\sqrt{\tau}}{\sqrt{N}} \right) \\ &= \frac{r\tau}{N} + \frac{\sigma\sqrt{\tau}}{\sqrt{N}} - \frac{1}{2} \left(\frac{r\tau}{N} + \frac{\sigma\sqrt{\tau}}{\sqrt{N}} \right)^2 + O\left(\frac{1}{N\sqrt{N}} \right) \\ &= \frac{r\tau}{N} + \frac{\sigma\tau}{\sqrt{N}} - \frac{\sigma^2\tau}{2N} + O\left(\frac{1}{N\sqrt{N}} \right). \end{aligned}$$

Similarly,

$$d = \frac{r\tau}{N} - \frac{\sigma\tau}{\sqrt{N}} - \frac{\sigma^2\tau}{2N} + O\left(\frac{1}{N\sqrt{N}} \right).$$

Therefore,

$$\begin{aligned}
\ln S_N &= \ln (S_0 u^{H_N} d^{T_N}) \\
&= \ln S_0 + H_N \ln u + T_N \ln d \\
&= \ln S_0 + H_N \left(\frac{r\tau}{N} + \frac{\sigma\sqrt{\tau}}{\sqrt{N}} - \frac{\sigma^2\tau}{2N} + O\left(\frac{1}{N\sqrt{N}}\right) \right) \\
&\quad + T_N \left(\frac{r\tau}{N} - \frac{\sigma\sqrt{\tau}}{\sqrt{N}} - \frac{\sigma^2\tau}{2N} + O\left(\frac{1}{N\sqrt{N}}\right) \right) \\
&= \sigma\sqrt{\tau} \frac{H_N - T_N}{\sqrt{N}} + \ln S_0 + \left(r - \frac{1}{2}\sigma^2 \right) \tau \frac{H_N + T_N}{N} \\
&\quad + \frac{H_N}{N} O\left(\frac{1}{\sqrt{N}}\right) + \frac{T_N}{N} O\left(\frac{1}{\sqrt{N}}\right) \\
&= \sigma\sqrt{\tau} \frac{M_N}{\sqrt{N}} + \ln S_0 + \left(r - \frac{1}{2}\sigma^2 \right) \tau + O\left(\frac{1}{\sqrt{N}}\right), \quad (8.4.28)
\end{aligned}$$

where we have used (8.2.10) and (8.2.9) in the last step. We define

$$Y_N = \ln S_0 + \left(r - \frac{1}{2}\sigma^2 \right) \tau + O\left(\frac{1}{\sqrt{N}}\right)$$

and note that

$$\lim_{N \rightarrow \infty} Y_N = \ln S_0 + \left(r - \frac{1}{2}\sigma^2 \right) \tau. \quad (8.4.29)$$

Furthermore, M_N is given by (8.2.3), where the random variables $X_1, X_2, \dots$ are independent and identically distributed with expected value 0 and variance 1. The Generalized Central Limit Theorem, Theorem 8.3.2, implies that for any bounded, continuous function $f(x)$ defined on $\mathbb{R}$,

$$\lim_{N \rightarrow \infty} \tilde{\mathbb{E}}[f(\ln S_N)] = \int_{-\infty}^{\infty} f\left(x\sigma\sqrt{\tau} + \ln S_0 + \left(r - \frac{1}{2}\sigma^2 \right) \tau\right) \varphi(x) dx. \quad (8.4.30)$$

To get the put payoff, we take

$$f(x) = (K - e^x)^+ \quad (8.4.31)$$

so that

$$f(\ln S_N) = (K - S_N)^+. \quad (8.4.32)$$

This function is continuous and bounded between 0 and K , and thus satisfies the conditions of the Generalized Central Limit Theorem. Using (8.4.25) and

(8.4.30), we see that the limit as $N \rightarrow \infty$ of the put price P_0 in (8.4.24) is

$$\begin{aligned}
& \lim_{N \rightarrow \infty} \frac{1}{(1 + \frac{r\tau}{N})^N} \tilde{\mathbb{E}}[(K - S_N)^+] \\
&= e^{-r\tau} \int_{-\infty}^{\infty} \left(K - \exp \left\{ x\sigma\sqrt{\tau} + \ln S_0 + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right\} \right)^+ \varphi(x) dx \\
&= \frac{e^{-r\tau}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(K - S_0 \exp \left\{ x\sigma\sqrt{\tau} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right\} \right)^+ \exp \left\{ -\frac{x^2}{2} \right\} dx.
\end{aligned} \tag{8.4.33}$$

It remains to compute the right-hand side of (8.4.33). We must first determine the values of x for which

$$K - S_0 \exp \left\{ x\sigma\sqrt{\tau} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right\} > 0, \tag{8.4.34}$$

so that the integrand on the right-hand side of (8.4.33) is not zero. Inequality (8.4.34) is equivalent to each of the following inequalities:

$$\begin{aligned}
K &> S_0 \exp \left\{ x\sigma\sqrt{\tau} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right\}, \\
\frac{K}{S_0} &> \exp \left\{ x\sigma\sqrt{\tau} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right\}, \\
\ln \frac{K}{S_0} &> x\sigma\sqrt{\tau} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \\
-\ln \frac{S_0}{K} - \left(r - \frac{1}{2}\sigma^2 \right) \tau &> x\sigma\sqrt{\tau}, \\
-\frac{1}{\sigma\sqrt{\tau}} \left[\ln \frac{S_0}{K} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right] &> x.
\end{aligned}$$

We define

$$d_2 = \frac{1}{\sigma\sqrt{\tau}} \left[\ln \frac{S_0}{K} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right]. \tag{8.4.35}$$

Then (8.4.34) is equivalent to

$$x < -d_2. \tag{8.4.36}$$

We only need to integrate the right-hand side of (8.4.33) over values of x satisfying (8.4.36); for other values of x , the integrand in (8.4.33) is zero.

Thus, the right-hand side of (8.4.33) is

$$\begin{aligned} & \frac{e^{-r\tau}}{\sqrt{2\pi}} \int_{-\infty}^{-d_2} \left(K - S_0 \exp \left\{ x\sigma\sqrt{\tau} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right\} \right) \exp \left\{ -\frac{x^2}{2} \right\} dx \\ &= \frac{e^{-r\tau}K}{\sqrt{2\pi}} \int_{-\infty}^{-d_2} e^{-x^2/2} dx - \frac{S_0}{\sqrt{2\pi}} \int_{-\infty}^{-d_2} \exp \left\{ -\frac{1}{2}(x - \sigma\sqrt{\tau})^2 \right\} dx. \end{aligned} \quad (8.4.37)$$

The first term on the right-hand side of (8.4.37) is $e^{-r\tau}KN(-d_2)$. In the second term, we make the change of variable $y = x - \sigma\sqrt{\tau}$. To find the upper limit of integration, we note that when $x = -d_2$,

$$\begin{aligned} y &= -d_2 - \sigma\sqrt{\tau} \\ &= -\frac{1}{\sigma\sqrt{\tau}} \left[\ln \frac{S_0}{K} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right] - \sigma\sqrt{\tau} \\ &= -\frac{1}{\sigma\sqrt{\tau}} \left[\ln \frac{S_0}{K} + \left(r - \frac{1}{2}\sigma^2 \right) \tau + \sigma^2\tau \right] \\ &= -\frac{1}{\sigma\sqrt{\tau}} \left[\ln \frac{S_0}{K} + \left(r + \frac{1}{2}\sigma^2 \right) \tau \right] = -d_1, \end{aligned}$$

where

$$d_1 = \frac{1}{\sigma\sqrt{\tau}} \left[\ln \frac{S_0}{K} + \left(r + \frac{1}{2}\sigma^2 \right) \tau \right]. \quad (8.4.38)$$

The second term on the right-hand side of (8.4.37) is

$$-\frac{S_0}{\sqrt{2\pi}} \int_{-\infty}^{-d_1} e^{-y^2/2} dy = -S_0N(-d_1).$$

We have thus determined that the right-hand side of (8.4.37) is

$$e^{-r\tau}KN(-d_2) - S_0N(-d_1).$$

This is the Black-Scholes price of a put. We summarize with a theorem.

Theorem 8.4.3 *The Black-Scholes price of a put with strike price K and time to expiration τ on a stock with volatility σ , obtained as the limit of the put price in a binomial model, is*

$$\lim_{N \rightarrow \infty} \frac{1}{(1 + \frac{r\tau}{N})^N} \tilde{\mathbb{E}}[(K - S_N)^+] = e^{-r\tau}KN(-d_2) - S_0N(-d_1), \quad (8.4.39)$$

where d_1 and d_2 are given by (8.4.38) and (8.4.35).

8.4.3 Black-Scholes price of a call

We derived the Black-Scholes price of a put rather than a call because the put pay-off function is bounded (see (8.4.31)), and this is required in order to use the Generalized Central Limit Theorem. The call payoff function is unbounded (in place of (8.4.31), for the call we would have $f(x) = (e^x - K)^+$, and this has limit ∞ as $x \rightarrow \infty$), and so we cannot directly apply the Generalized Central Limit Theorem to the call pricing problem. However, we can use put-call parity to derive the call price from the put price, and we do that now.

Theorem 8.4.4 *The Black-Scholes price of a call with strike price K and time to expiration τ on a stock with volatility σ , obtained as the limit of the call price in a binomial model, is*

$$\lim_{N \rightarrow \infty} \frac{1}{(1 + \frac{r\tau}{N})^N} \tilde{\mathbb{E}}[(S_N - K)^+] = S_0 N(d_1) - e^{-r\tau} K N(d_2), \quad (8.4.40)$$

where d_1 and d_2 are given by (8.4.38) and (8.4.35).

PROOF: Let

$$C_0 = \frac{1}{(1 + \frac{r\tau}{N})^N} \tilde{\mathbb{E}}[(S_N - K)^+],$$

and recall from (8.4.24) the price of the put

$$P_0 = \frac{1}{(1 + \frac{r\tau}{N})^N} \tilde{\mathbb{E}}[(K - S_N)^+].$$

Because

$$(S_N - K)^+ - (K - S_N)^+ = S_N - K,$$

we have

$$C_0 - P_0 = \frac{1}{(1 + \frac{r\tau}{N})^N} \tilde{\mathbb{E}}[S_N - K].$$

But $\frac{S_n}{(1 + \frac{r\tau}{N})^n}$, $n = 0, 1, \dots, N$, is a martingale under the risk-neutral probability measure, so

$$C_0 - P_0 = S_0 - \frac{K}{(1 + \frac{r\tau}{N})^N}.$$

From this equation, Theorem 8.4.3, and equation (8.4.25), we have

$$\begin{aligned} \lim_{N \rightarrow \infty} C_0 &= \lim_{N \rightarrow \infty} P_0 + S_0 - \lim_{N \rightarrow \infty} \frac{K}{(1 + \frac{r\tau}{N})^N} \\ &= e^{-r\tau} K N(-d_2) - S_0 N(-d_1) + S_0 - e^{-r\tau} K \\ &= S_0(1 - N(-d_1)) - e^{-r\tau} K(1 - N(-d_2)). \end{aligned} \quad (8.4.41)$$

Finally, we use (8.3.20) to obtain (8.4.40) from (8.4.41).

8.4.4 Summary of formulas

For future reference, we record here the Black-Scholes formulas for the price of a call and a put. In these formulas, s is the price of the underlying stock at the time of pricing and τ is the time until expiration of the option. If the option expires at T and the time of pricing is t , where $0 \leq t < T$, then

$$\tau = T - t, \quad (8.4.42)$$

and s is the stock price at time t . The variables d_1 and d_2 depend on both τ and s , and we indicate that explicitly in (8.4.43)–(8.4.46) below.

The Black-Scholes price of a call is

$$c(t, s) = sN(d_1(\tau, s)) - e^{-r\tau}KN(d_2(\tau, s)), \quad (8.4.43)$$

and the Black-Scholes price of a put is

$$p(t, s) = e^{-r\tau}KN(-d_2(\tau, s)) - sN(-d_1(\tau, s)), \quad (8.4.44)$$

where

$$d_1(\tau, s) = \frac{1}{\sigma\sqrt{\tau}} \left[\ln \frac{s}{K} + \left(r + \frac{1}{2}\sigma^2 \right) \tau \right], \quad (8.4.45)$$

$$d_2(\tau, s) = \frac{1}{\sigma\sqrt{\tau}} \left[\ln \frac{s}{K} + \left(r - \frac{1}{2}\sigma^2 \right) \tau \right]. \quad (8.4.46)$$

The call and put prices are related by *put-call parity* (see (8.3.20)):

$$\begin{aligned} c(t, s) - p(t, s) &= s [N(d_1(\tau, s)) + N(-d_1(\tau, s))] \\ &\quad - e^{-r\tau}K [N(d_2(\tau, s)) + N(-d_2(\tau, s))] \\ &= s - e^{-r\tau}K. \end{aligned} \quad (8.4.47)$$

8.5 Black-Scholes Partial Differential Equation

The Black-Scholes functions $c(t, s)$ and $p(t, s)$ satisfy the *Black-Scholes partial differential equation*

$$v_t(t, s) + rsv_s(t, s) + \frac{1}{2}\sigma^2s^2v_{ss}(t, s) = rv(t, s). \quad (8.5.1)$$

Here we use the notation $v_t(t, s)$ for the partial derivative $\frac{\partial}{\partial t}v(t, s)$, we use $v_s(t, s)$ to denote $\frac{\partial}{\partial s}v(t, s)$, and we use $v_{ss}(t, s)$ to denote the second-order partial derivative $\frac{\partial^2}{\partial s^2}v(t, s)$. The Black-Scholes partial differential equation plays an important role in the construction of replicating portfolios.

In this section we begin with the backward recursion developed for the binomial model and pass to the limit to derive the Black-Scholes partial differential equation. This equation is satisfied by both the price of a call and the price of a put.

8.5.1 Fixing a time t

Consider a time interval $[0, \tau]$, which we divide into N periods, each of length $\frac{\tau}{N}$. The annual interest rate is r and the stock volatility is σ . The parameters in this N -period model depend on N , and we use a subscript N to remind us of this fact. The up and down factors are

$$u_N = 1 + \frac{r\tau}{N} + \frac{\sigma\sqrt{\tau}}{\sqrt{N}}, \quad d_N = 1 + \frac{r\tau}{N} - \frac{\sigma\sqrt{\tau}}{\sqrt{N}}.$$

The interest rate per period is $r_N = \frac{r\tau}{N}$. The risk-neutral probabilities are

$$\tilde{p} = \frac{1 + r_N - d_N}{u_N - d_N} = \frac{1}{2}, \quad \tilde{q} = \frac{u_N - 1 - r_N}{u_N - d_N} = \frac{1}{2}.$$

When we divide the time interval $[0, \tau]$ into N periods, we need to be clear about what we mean by “time.” If $\tau = \frac{1}{4}$, then in a 50-period model, the beginning of period zero will be time 0 and the beginning of period 25 will be at time $\frac{1}{8}$. In a 100-period model with $\tau = \frac{1}{4}$, time $\frac{1}{8}$ is at the beginning of period 50. We shall speak of “the beginning of period 25” rather than “time 25.” When we say “time,” we mean a number between 0 and τ , not a value of n in the binomial model.

The backward recursion for the N -period model is

$$v_{n,N}(s) = \frac{1}{1 + r_N} \left[\frac{1}{2}v_{n+1,N}(u_N s) + \frac{1}{2}v_{n+1,N}(d_N s) \right]. \quad (8.5.2)$$

Here we denote by $v_{n,N}(s)$ the value of an option in the N -period model at the beginning of period n if the stock price at that time is s . We thus denote by $v_{N,N}(s)$ the payoff of the option at time τ . For a call, $v_{N,N}(s) = (s - K)^+$.

We fix a time t and consider the value of an option at this time in a sequence of binomial models. We do this by letting both n and N go to

infinity in such a way that $t = \frac{n\tau}{N}$ remains fixed. For example, if $\tau = \frac{1}{4}$ and we want to consider the time $t = \frac{1}{8}$, in the 50 period model we choose $n = 25$ and in the 100 period model we choose $n = 50$. For $t = \frac{1}{8}$, we always choose $n = \frac{N}{2}$ and let both n and N go to infinity. For $t = \frac{1}{8}$ we would not consider odd values of N , since there is no period beginning at time $t = \frac{1}{8}$ when we divide $[0, \frac{1}{4}]$ into an odd number of periods.

We denote by $v(t, s)$ the limit of $v_{n,N}(s)$ as $N \rightarrow \infty$ where we also let $n \rightarrow \infty$ so that the ratio $\frac{n\tau}{N}$ is always equal to t . If we are considering a call, then $v(t, s)$ will be the Black-Scholes call price $c(t, s)$ given by (8.4.43); if we are considering a put, then $v(t, s)$ will be the put price $p(t, s)$ given by (8.4.44).

When $t = \frac{n\tau}{N}$, then period $n + 1$ begins at time $t + \frac{\tau}{N}$. We replace $v_{n,N}(s)$ in (8.5.2) by the limit $v(t, s)$, replace $v_{n+1,N}(u_N s)$ by $v(t + \frac{\tau}{N}, u_N s)$, and replace $v_{n+1,N}(d_N s)$ by $v(t + \frac{\tau}{N}, d_N s)$. We are replacing one set of terms by other terms that are only approximately the same. This introduces an error, and the size of that error depends on the value of N . In fact, the error is not larger than a constant times $\frac{1}{N\sqrt{N}}$, so we say the error is $O(\frac{1}{N\sqrt{N}})$. It is difficult to show that this is the size of the error, and we do not attempt it here.

8.5.2 Expansion of the backward recursion formula

We write (8.5.2) as

$$v_{n+1,N}(u_N s) + v_{n+1,N}(d_N s) - 2(1 + r_N)v_{n,N}(s) = 0$$

and make the replacements just described. Because of the error introduced, we no longer have 0 on the right-hand side of the equation, but instead have $O(\frac{1}{N\sqrt{N}})$. Indeed,

$$v\left(t + \frac{\tau}{N}, u_N s\right) + v\left(t + \frac{\tau}{N}, d_N s\right) - 2(1 + r_N)v(t, s) = O\left(\frac{1}{N\sqrt{N}}\right). \quad (8.5.3)$$

We rearrange the left-hand side of (8.5.3) to obtain

$$\begin{aligned} & \left[v\left(t + \frac{\tau}{N}, u_N s\right) - v(t, u_N s) \right] + [v(t, u_N s) - v(t, s)] \\ & + \left[v\left(t + \frac{\tau}{N}, d_N s\right) - v(t, d_N s) \right] + [v(t, d_N s) - v(t, s)] \\ & - 2r_N v(t, s) = O\left(\frac{1}{N\sqrt{N}}\right). \end{aligned} \quad (8.5.4)$$

We now apply Taylor's Theorem. In (8.5.5) and (8.5.6) below, we hold the second argument of v fixed and apply Taylor's Theorem to the variable t . In (8.5.7) and (8.5.8), we hold the first argument of v fixed and apply Taylor's Theorem to the variable s . We have the four expansions

$$v\left(t + \frac{\tau}{N}, u_N s\right) = v(t, u_N s) + v_t(t, u_N s) \frac{\tau}{N} + O\left(\frac{1}{N^2}\right), \quad (8.5.5)$$

$$v\left(t + \frac{\tau}{N}, d_N s\right) = v(t, d_N s) + v_t(t, d_N s) \frac{\tau}{N} + O\left(\frac{1}{N^2}\right), \quad (8.5.6)$$

$$\begin{aligned} v(t, u_N s) &= v(t, s) + v_s(t, s)(u_N - 1)s + \frac{1}{2}v_{ss}(t, s)(u_N - 1)^2 s^2 \\ &\quad + O((u_N - 1)^3 s^3), \end{aligned} \quad (8.5.7)$$

$$\begin{aligned} v(t, d_N s) &= v(t, s) + v_s(t, s)(d_N - 1)s + \frac{1}{2}v_{ss}(t, s)(d_N - 1)^2 s^2 \\ &\quad + O((d_N - 1)^3 s^3). \end{aligned} \quad (8.5.8)$$

Because $u_N - 1 = O(\frac{1}{\sqrt{N}})$ and $d_N - 1 = O(\frac{1}{\sqrt{N}})$, the $O(\cdot)$ terms appearing on the right-hand side of (8.5.7) and (8.5.8) are $O(\frac{1}{N\sqrt{N}})$. Furthermore,

$$(u_N - 1)^2 = \frac{\sigma^2 \tau}{N} + O\left(\frac{1}{N\sqrt{N}}\right), \quad (8.5.9)$$

$$(d_N - 1)^2 = \frac{\sigma^2 \tau}{N} + O\left(\frac{1}{N\sqrt{N}}\right). \quad (8.5.10)$$

Using (8.5.5), (8.5.7), (8.5.6) and (8.5.8), in that order, we see that the first four terms on the left-hand side of (8.5.4) are

$$v\left(t + \frac{\tau}{N}, u_N s\right) - v(t, u_N s) = \frac{\tau}{N} v_t(t, u_N s) + O\left(\frac{1}{N^2}\right) \quad (8.5.11)$$

$$\begin{aligned} v(t, u_N s) - v(t, s) &= \left(\frac{r\tau}{N} + \frac{\sigma\sqrt{\tau}}{\sqrt{N}}\right) s v_s(t, s) + \frac{\sigma^2 \tau s^2}{2N} v_{ss}(t, s) \\ &\quad + O\left(\frac{1}{N\sqrt{N}}\right), \end{aligned} \quad (8.5.12)$$

$$v\left(t + \frac{\tau}{N}, d_N s\right) - v(t, d_N s) = \frac{\tau}{N} v_t(t, d_N s) + O\left(\frac{1}{N^2}\right), \quad (8.5.13)$$

$$\begin{aligned} v(t, d_N s) - v(t, s) &= \left(\frac{r\tau}{N} - \frac{\sigma\sqrt{\tau}}{\sqrt{N}}\right) s v_s(t, s) + \frac{\sigma^2 \tau s^2}{2N} v_{ss}(t, s) \\ &\quad + O\left(\frac{1}{N\sqrt{N}}\right). \end{aligned} \quad (8.5.14)$$

We substitute these expressions into (8.5.4), and we do not write any $O(\frac{1}{N^2})$ terms because an $O(\frac{1}{N\sqrt{N}})$ term is present. This yields

$$\begin{aligned} \frac{\tau}{N}v_t(t, u_Ns) + \frac{\tau}{N}v_t(t, d_Ns) + \frac{2r\tau}{N}sv_s(t, s) + \frac{\sigma^2\tau s^2}{N}v_{ss}(t, s) - \frac{2r\tau}{N}v(t, s) \\ = O\left(\frac{1}{N\sqrt{N}}\right). \end{aligned} \quad (8.5.15)$$

We multiply (8.5.15) by $\frac{N}{2\tau}$ to obtain the *expanded backward recursion formula*

$$\frac{v_t(t, u_Ns) + v_t(t, d_Ns)}{2} + rsv_s(t, s) + \frac{1}{2}\sigma^2s^2v_{ss}(t, s) - rv(t, s) = O\left(\frac{1}{\sqrt{N}}\right). \quad (8.5.16)$$

8.5.3 Black-Scholes partial differential equation

Equation (8.5.16) is a Taylor expansion of the binomial backward recursion formula (8.5.2). As a last step in this discussion, we let $N \rightarrow \infty$. We observe that $\lim_{N \rightarrow \infty} u_N = 1$ and $\lim_{N \rightarrow \infty} d_N = 1$, so

$$\lim_{N \rightarrow \infty} \frac{v_t(t, u_Ns) + v_t(t, d_Ns)}{2} = \frac{v_t(t, s) + v_t(t, s)}{2} = v_t(t, s).$$

As $N \rightarrow \infty$, the right-hand side of (8.5.16) converges to zero. Thus, taking the limit in (8.5.16), we obtain

$$v_t(t, s) + rsv_s(t, s) + \frac{1}{2}\sigma^2s^2v_{ss}(t, s) - rv(t, s) = 0. \quad (8.5.17)$$

This is the Black-Scholes partial differential equation (8.5.1). Because we have derived (8.5.1) from the backward recursion formula (8.5.2) that is satisfied by both the put and the call prices, this equation is satisfied by both $c(t, s)$ given by (8.4.43) and $p(t, s)$ given by (8.4.44).

8.5.4 Delta hedging

In the binomial model, we can replicate an option by trading in the underlying stock and borrowing or investing in the money market account as necessary to finance this. If the price of a share of stock at the beginning of

period n is s , then the number of shares of the stock held by the replicating portfolio from the beginning to the end of period n is

$$\Delta_n(s) = \frac{v_{n+1,N}(u_N s) - v_{n+1,N}(d_N s)}{u_N s - d_N s} = \frac{v_{n+1,N}(u_N s) - v_{n+1,N}(d_N s)}{(u_N - d_N)s}. \quad (8.5.18)$$

For large values of N , it is possible to replace (8.5.18) by a simpler formula by determining the limit in (8.5.18) as $N \rightarrow \infty$.

Again we fix t and let N and n both approach infinity so that $t = \frac{n\tau}{N}$ remains constant. We first observe that

$$(u_N - d_N)s = \frac{2\sigma s\sqrt{\tau}}{\sqrt{N}}. \quad (8.5.19)$$

In (8.5.18) we substitute (8.5.19), we replace $v_{n+1,N}(u_N s)$ by $v(t + \frac{\tau}{N}, u_N s)$, and we replace $v_{n+1,N}(d_N s)$ by $v(t + \frac{\tau}{N}, d_N s)$. This introduces an error $O(\frac{1}{\sqrt{N}})$, so that (8.5.18) becomes

$$\Delta(t, s) = \frac{\sqrt{N}}{2\sigma s\sqrt{\tau}} \left[v\left(t + \frac{\tau}{N}, u_N s\right) - v\left(t + \frac{\tau}{N}, d_N s\right) \right] + O\left(\frac{1}{\sqrt{N}}\right). \quad (8.5.20)$$

We call the expression in (8.5.20) $\Delta(t, s)$ rather than Δ_n because as $N \rightarrow \infty$ (and along with it, $n \rightarrow \infty$), this expression will depend on the time t and the underlying stock price s at that time but not on the period n .

Using (8.5.11) and (8.5.13) in the second equality below, we have

$$\begin{aligned} \Delta(t, s) &= \frac{\sqrt{N}}{2\sigma s\sqrt{\tau}} \left[\left(v\left(t + \frac{\tau}{N}, u_N s\right) - v(t, u_N s) \right) + \left(v(t, u_N s) - v(t, d_N s) \right) \right. \\ &\quad \left. - \left(v\left(t + \frac{\tau}{N}, d_N s\right) - v(t, d_N s) \right) \right] + O\left(\frac{1}{\sqrt{N}}\right) \\ &= \frac{\sqrt{N}}{2\sigma s\sqrt{\tau}} \left[\frac{\tau}{N} v_t(t, u_N s) + v(t, u_N s) - v(t, d_N s) - \frac{\tau}{N} v_t(t, d_N s) \right. \\ &\quad \left. + O\left(\frac{1}{N^2}\right) \right] + O\left(\frac{1}{\sqrt{N}}\right) \\ &= \frac{\sqrt{N}}{2\sigma s\sqrt{\tau}} [v(t, u_N s) - v(t, d_N s)] + O\left(\frac{1}{\sqrt{N}}\right). \end{aligned} \quad (8.5.21)$$

From (8.5.7), (8.5.8), (8.5.9), (8.5.10), and (8.5.19), we have

$$\begin{aligned} v(t, u_N s) - v(t, d_N s) &= v_s(t, s)(u_N - d_N)s + O\left(\frac{1}{N}\right) \\ &= \frac{2\sigma s\sqrt{\tau}}{\sqrt{N}}v_s(t, s) + O\left(\frac{1}{N}\right). \end{aligned} \quad (8.5.22)$$

Substitution of (8.5.22) into (8.5.21) yields

$$\Delta(t, s) = v_s(t, s) + O\left(\frac{1}{\sqrt{N}}\right). \quad (8.5.23)$$

Letting $N \rightarrow \infty$, we see that at each time t , if the stock price is s , the replicating portfolio should hold

$$\Delta(t, s) = v_s(t, s). \quad (8.5.24)$$

This is called the *delta* of the option. When we hold $\Delta(t, s)$ shares of the underlying stock at time t , we are said to be “delta hedging.”

For a call option, whose price is given by (8.4.43), the delta of the option is (see Exercise 8.1(ii))

$$c_s(t, s) = N(d_1(\tau, s)). \quad (8.5.25)$$

For the put option, whose price is given by (8.4.44), the delta of the option is (see Exercise 8.5)

$$p_s(t, s) = N(d_1(\tau, s)) - 1 = -N(-d_1(\tau, s)). \quad (8.5.26)$$

8.5.5 Delta, gamma and theta

Consider an agent who sells a call at time zero for $c(0, S(0))$, where $S(0)$ denotes the stock price at time zero. The agent is then obligated to deliver $(S(\tau) - K)^+$ at the expiration time τ , where K is the strike price of the call and $S(\tau)$ is the stock price at the expiration time τ . In an N -period binomial model, this agent can take the capital $c(0, S(0))$ and invest in the underlying stock, financing this by investing or borrowing from the money market account, in order to have a portfolio valued at $(S(\tau) - K)^+$ at time τ . This replicating portfolio is defined by equation (8.5.18). In particular, at the beginning of each period n , the agent should take a position of Δ_n shares of stock, where Δ_n is given by (8.5.18).

Let us fix a time t between 0 and T . In the N -period model over the interval $[0, T]$, we choose n so that $t = \frac{n\tau}{N}$. (We actually consider only times t that are rational numbers. Then we can always find pairs n and N such that $t = \frac{n\tau}{N}$, and we can do this so that both N and n go to infinity.) When $t = \frac{n\tau}{N}$ and N is large, $\Delta_n(s)$ given by (8.5.18) is nearly the same as $\Delta(t, s)$ given by (8.5.24) with $v(t, s)$ replaced by the call price $c(t, s)$.

Suppose the stock price at time t is s_1 . We wish to understand the cash flow incurred by an agent who is short one call option at time t , is long $\Delta(t, s_1) = c_s(t, s_1)$ shares of the underlying stock, and has a position in the money market account to finance her trading in the underlying stock. We call such an agent a *short call hedger*.

Note that at time t , the time to expiration of the call is $T - t$. Therefore,

$$c(t, s_1) = s_1 N(d_1) - e^{-r(T-t)} K N(d_2),$$

where

$$\begin{aligned} d_1 &= \frac{1}{\sigma\sqrt{T-t}} \left[\ln \frac{s_1}{K} + \left(r + \frac{1}{2}\sigma^2 \right) (T-t) \right], \\ d_2 &= \frac{1}{\sigma\sqrt{T-t}} \left[\ln \frac{s_1}{K} + \left(r - \frac{1}{2}\sigma^2 \right) (T-t) \right]. \end{aligned}$$

The short call hedger has constructed a replicating portfolio whose value at time t is the same as the value of the call, $c(t, s_1)$. The agent holds $\Delta(t, s_1) = c_s(t, s_1)$ shares of stock. According to Exercise 8.1(ii), $\Delta(t, s_1) = c_s(t, s_1) = N(d_1)$, and so the value of the stock held by the agent is $s_1 N(d_1)$. The difference

$$\begin{aligned} c(t, s_1) - s_1 N(d_1) &= (s_1 N(d_1) - e^{-r(T-t)} K N(d_2)) - s_1 N(d_1) \\ &= -e^{-r(T-t)} K N(d_2) \end{aligned}$$

is invested in the money market account. This is negative, which means that the agent has borrowed money from the money market account. The agent hedging the short-call position is short one call, which is valued at $c(t, s_1)$. If we include this short call position, then the total wealth of the agent is

$$\begin{aligned} & -\text{Value of call} + \text{Value of stock held} + \text{Cash in money market account} \\ &= -c(t, s_1) + s_1 N(d_1) - e^{-r(T-t)} K N(d_2) \\ &= 0. \end{aligned} \tag{8.5.27}$$

The portfolio in (8.5.27), holding the one call, some stock, and having a position in the money market account, is said to be *delta-neutral* because its derivative with respect to the underlying stock price is zero. More specifically, if the agent does not change her positions in the stock and money market account, i.e., if $N(d_1)$ and $-e^{-r(T-t)}KN(d_2)$ are held constant, then the only dependence on the stock price in the total wealth of the agent appears in the term $c(t, s_1)$ and the s_1 in the term $s_1N(d_1)$. If the stock price moves to s , and does so quickly so we can neglect the interest accumulation in the money market account, the total value of the agent's wealth will change to

$$-c(t, s) + sN(d_1) - e^{-r(T-t)}KN(d_2).$$

The derivative of this with respect to s , evaluated at $s = s_1$, is

$$\left. \frac{\partial}{\partial s} [-c(t, s) + sN(d_1) - e^{-r(T-t)}KN(d_2)] \right|_{s=s_1} = -c_s(t, s_1) + N(d_1) = 0. \quad (8.5.28)$$

In general, the *delta* of a portfolio is its derivative with respect to the stock price when all the positions are held constant. The portfolio in (8.5.27) not only has value zero, it also has a zero delta, which is why the portfolio is said to be delta-neutral. A portfolio that has positive delta is said to be *long delta*. An example of such a portfolio is a *naked call*, a portfolio that holds one call and has zero position in the underlying stock. (The position in the money market account is irrelevant in determining the delta of a portfolio because this is not affected by stock price movements.) The delta of a naked call is $N(d_1)$. A portfolio that has a negative delta is said to be *short delta*.

We now consider an agent who is hedging a long-call position. Again the time is t and the stock price at time t is assumed to be s_1 . This agent has the opposite position of the agent who is short call hedger. In particular, the long-call hedger owns one call, is short $\Delta(t, s_1) = c_s(t, s_1) = N(d_1)$ shares of stock, and has invested $e^{-r(T-t)}N(d_1)$ in the money market account. This agent has total wealth

$$\begin{aligned} & \text{Value of call} + \text{Value of stock held} + \text{Cash in money market account} \\ &= c(t, s_1) - sN(d_1) + e^{-r(T-t)}KN(d_2) \\ &= 0. \end{aligned} \quad (8.5.29)$$

Furthermore, the delta of the portfolio in (8.5.29) is zero:

$$\left. \frac{\partial}{\partial s} [c(t, s) - sN(d_1) + e^{-r(T-t)}KN(d_2)] \right|_{s=s_1} = c_s(t, s_1) - N(d_1) = 0. \quad (8.5.30)$$

We want to examine the *second derivative* in (8.5.30), the so-called *gamma* of the portfolio. This second derivative is

$$\frac{\partial^2}{\partial s^2} [c(t, s) + sN(d_1) - e^{-r(T-t)}KN(d_2)] \Big|_{s=s_1} = c_{ss}(t, s_1). \quad (8.5.31)$$

It was computed in Exercise 8.1(iii) and is

$$c_{ss}(t, s_1) = \frac{1}{\sigma s_1 \sqrt{T-t}} N'(d_1), \quad (8.5.32)$$

which is positive. Indeed, the call price $c(t, s)$ is a convex function of the stock price s . Consequently, we say the portfolio in (8.5.29) is *long gamma*.

Let us consider what happens to the value of the portfolio in (8.5.29) if the stock moves up from s_1 to a value $s_2 > s_1$. We assume this happens quickly so we may neglect the interest accumulation in the money market account. The value of the agent's portfolio after the stock price move is

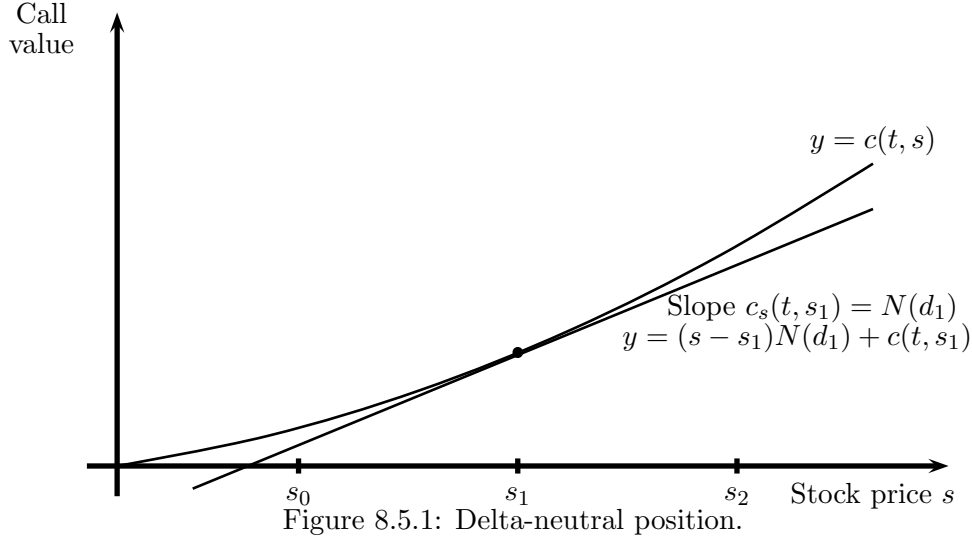
$$\begin{aligned} & c(t, s_2) - s_2 N(d_1) + e^{-r(T-t)}KN(d_2) \\ &= c(t, s_2) - (s_2 - s_1)N(d_1) - [s_1 N(d_1) - e^{-r(T-t)}KN(d_2)] \\ &= c(t, s_2) - (s_2 - s_1)N(d_1) - c(t, s_1). \end{aligned} \quad (8.5.33)$$

This is the difference between the curve $y = c(t, s)$ and the tangent line $y = (s - s_1)N(d_1) + c(t, s_1)$ evaluated at $s = s_2$; see Figure 8.5.1. Because the curve has positive second derivative (see (8.5.32)), the expression in (8.5.33) is positive. When the stock price rises from s_1 to s_2 , the value of the call held by the long call hedger rises more than the loss due to the short stock position in his portfolio. If all other effects are ignored (they cannot be, as we see below), then long gamma is a desirable property for a portfolio when the stock rises.

But what happens to the value of the long call hedger's portfolio if the stock falls? Once again we discover that he makes money. Suppose the stock moves down from s_1 to a value $s_0 < s_1$. Again, we assume this happens quickly. The value of the agent's portfolio after the stock price move is

$$\begin{aligned} & c(t, s_0) - s_0 N(d_1) + e^{-r(T-t)}KN(d_2) \\ &= c(t, s_0) - (s_0 - s_1) - [s_1 N(d_1) - e^{-r(T-t)}KN(d_2)] \\ &= c(t, s_0) - (s_0 - s_1)N(d_1) - c(t, s_1). \end{aligned} \quad (8.5.34)$$

Again the value of the portfolio is the difference between the curve $y = c(t, s)$ and the tangent line $y = (s - s_1)N(d_1) + c(t, s_1)$, this time evaluated at $s = s_0$.



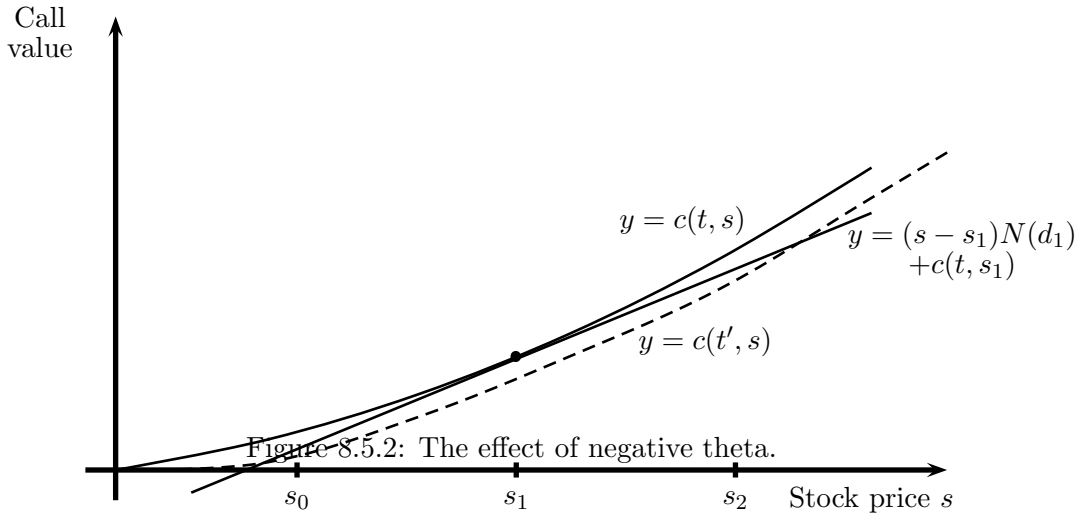
Once again, this is positive because the call price is a convex function of the stock price; see Figure 8.5.1.

This discussion makes it appear that the long call hedger has created an arbitrage. He has a portfolio that is long one call, short some amount of stock, and has a positive amount of cash in the money market. Because this portfolio has value zero at time t (see (8.5.29)), he can set it up at time t with zero initial investment. Then, regardless of whether the stock goes up or down, the value of the portfolio appears to go up. There are, however, two factors that have been neglected. One is that it takes some time for the stock to move, and during that time, the cash in the money market account earns interest. This works to the advantage of the long call hedger. The other factor is that during the time it takes for the stock price to move, the value of the call tends to go down. This is because *theta*, the partial derivative of the call with respect to time, is negative. This partial derivative is (see Exercise 8.1(iv))

$$c_t(t, s_1) = -rKe^{-r(T-t)}N(d_2) - \frac{\sigma s_1}{2\sqrt{T-t}}N'(d_1),$$

and both $N(d_2)$ and $N'(d_1)$ are positive.

Because $c_t(t, s_1)$ is negative, if the stock price stays at s_1 , then the value of the call decreases as time advances. However, the value of the stock may move up as time advances, and because the value of the call increases with increasing stock price, we may in fact see the value of the call increase as time



moves forward and the stock price moves up. Of course, if stock price moves down as time moves forward, the value of the call will definitely decrease. In any event, when we compare the value of the portfolio in (8.5.29) with its value after the stock price changes, we should also take into account that time advances to some new time t' , and $c(t', s_1) < c(t, s_1)$. Figure 8.5.1 does not tell the whole story, because the curve $y = c(t, s)$ that is used when the stock price is s_1 lies above the curve that should be used to price the call after the stock price moves.

A more complete description of the situation is shown in Figure 8.5.2. When time moves forward from t to t' , the call price function drops from the solid curve $y = c(t, s)$ to the dashed curve $y = c(t', s)$. Along with this, the stock price moves, perhaps up to some value such as s_2 or perhaps down to some value such as s_0 . The long call hedger gains because of convexity, loses because of negative theta, and gains in a way not shown in Figure 8.5.2 because he has cash invested in the money market account. For a perfect hedge, these three effects cancel. That is exactly what happens when we construct a replicating portfolio in a binomial model. Note, however, that the replicating portfolio in a binomial model calls for an adjustment of the position in the stock and money market account at the beginning of each period. Once we have let the number of periods go to infinity to obtain the Black-Scholes formula, it will be necessary to continuously adjust the position in the stock and the money market account to get the replicating portfolio to work. *Stochastic calculus* is the branch of mathematics that permits us to analyze this situation.

8.6 Volatility and Return

Inherent in the Black-Scholes option pricing formula are two profound observations. The first is that the price of a call or put option on a stock does not depend on the expected return of the stock. As an extreme case of this, we note that the value of an option to buy a share of stock at a future date for $K = 0$ should be the price of the stock today. This is the value given by the Black-Scholes formula as $K \downarrow 0$ in (8.4.43), and it does not depend on the expected return of the stock. If anyone argues that the option should be worth more than today's stock price because the stock has a high expected return, and if this person is willing to pay more than today's stock price for this option, then you should sell him as many of these options as he will buy. With the proceeds of sale, you should buy one share of stock for each option sold and pocket the remaining income. You then deliver the stock shares to the option purchaser at the future time, which fulfills your obligation, and you still have money in your pocket. This is an arbitrage.

When the strike price of a call option is not zero and someone argues that the option price should be high because the stock has a high expected return, it is not so easy to realize an arbitrage of the kind just discussed. After you sell options to this person, you need to actively trade in the stock and the money market account in order to fulfill your obligation to the option purchaser. To determine how to do this trading, you need a model for the stock price movements. The Black-Scholes formula is based on a particular model.

The second profound observation inherent in the Black-Scholes formula is the critical role played by *volatility*. Volatility is sometimes confused with the variance of the growth rate of a stock. In fact, volatility is a deeper concept than variance. It measures how much movement one observes in the stock price process. In these notes, we will assume that the volatility is constant over time and is not random. These assumptions can be relaxed, but we cannot do that without using stochastic calculus.

8.6.1 Rate of growth

We consider a time interval of one trading day and divide this interval into K periods so that each period has length equal to a fraction $\frac{1}{K}$ of the trading day. We build a binomial model in which we have a *money market account* that pays interest $\frac{rD}{K}$ per period. We also have a *stock*, whose price moves

up or down in each period, with up and down factors

$$u = 1 + \frac{r_D}{K} + \frac{\sigma_D}{\sqrt{K}}, \quad d = 1 + \frac{r_D}{K} - \frac{\sigma_D}{\sqrt{K}}. \quad (8.6.1)$$

Here we denote by r_D the *daily interest rate* and by σ_D the *daily volatility*. We will discuss how these are related to the annual interest rate and the annual volatility in Subsection 8.6.3. For the moment, we note only that r_D can be determined in practice from announced money market rates, whereas σ_D is more difficult to determine (see Subsection 8.6.2). For the following discussion, we require only that σ_D is positive and chosen so that the down factor d is also positive. This is done so the no-arbitrage condition

$$d < 1 + \frac{r_D}{K} < u$$

is satisfied.

As usual, we toss a coin to determine if the stock price moves up or down in each period. However, we do not assume this coin is fair (which would be the case for the risk-neutral probabilities). Let p be the probability of a head on each toss and $q = 1 - p$ the probability of a tail. We assume that we have observed the stock price for some time, we have estimated the daily return, and we want to choose p (and hence $q = 1 - p$) in order to get our model to match what we have observed.

There is more than one way to measure daily return. The two we consider here are $\ln \mathbb{E}[\frac{S_K}{S_0}]$ and $\mathbb{E}[\ln \frac{S_K}{S_0}]$. We call $\mathbb{E}[\frac{S_K}{S_0}]$ the *expected growth* per day of the stock, so $\ln \mathbb{E}[\frac{S_K}{S_0}]$ is the *rate of expected growth*. On the other hand, $\ln \frac{S_K}{S_0}$ is the *rate of growth* of the stock, so $\mathbb{E}[\ln \frac{S_K}{S_0}]$ is the *expected rate of growth*. Because of Jensen's equality and the fact that $\ln$ is concave ($-\ln$ is convex), we have

$$\ln \mathbb{E} \left[\frac{S_K}{S_0} \right] \geq \mathbb{E} \left[\ln \frac{S_K}{S_0} \right]. \quad (8.6.2)$$

For major stocks, trading occurs frequently and so K should be a large number. Therefore, we shall compute the two sides of (8.6.2) as the number of periods K into which we divide one trading day approaches infinity. The inequality (8.6.2) will still hold in the limit, i.e., we will have

$$\lim_{K \rightarrow \infty} \ln \mathbb{E} \left[\frac{S_K}{S_0} \right] \geq \lim_{K \rightarrow \infty} \mathbb{E} \left[\ln \frac{S_K}{S_0} \right]. \quad (8.6.3)$$

For $k = 1, 2, \dots, K$, we have

$$\mathbb{E} \left[\frac{S_k}{S_{k-1}} \right] = pu + qd = 1 + \frac{r_D}{K} + \frac{(p-q)\sigma_D}{\sqrt{K}}. \quad (8.6.4)$$

If $0 < q < \frac{1}{2} < p < 1$, which is the normal situation, the right-hand side of (8.6.4) is greater than $1 + \frac{r_D}{K}$, the result we would get using the risk-neutral probabilities. Indeed, under the actual probabilities, we would expect to have

$$\mathbb{E} \left[\frac{S_k}{S_{k-1}} \right] = 1 + \frac{\mu_D}{K}, \quad (8.6.5)$$

where μ_D is some number greater than r_D . To achieve this, when there are K periods per day, we choose

$$p = \frac{1}{2} + \frac{\mu_D - r_D}{2\sigma_D\sqrt{K}}, \quad q = \frac{1}{2} - \frac{\mu_D - r_D}{2\sigma_D\sqrt{K}}, \quad (8.6.6)$$

so that

$$p - q = \frac{\mu_D - r_D}{\sigma_D\sqrt{K}} \quad (8.6.7)$$

and then (8.6.4) becomes (8.6.5).

It is important to note that we have not changed the up and down factors. They are still given by (8.6.1). The binomial tree used with the risk-neutral probabilities is the same binomial tree used with probabilities p and q given by (8.6.6). Only the probabilities have changed.

With p and q given by (8.6.6), we have

$$\begin{aligned} \mathbb{E} \left[\frac{S_K}{S_0} \right] &= \mathbb{E} \left[\frac{S_1}{S_0} \cdot \frac{S_2}{S_1} \cdots \frac{S_{K-1}}{S_{K-2}} \cdot \frac{S_K}{S_{K-1}} \right] \\ &= \mathbb{E} \left[\frac{S_1}{S_0} \right] \cdot \mathbb{E} \left[\frac{S_2}{S_1} \right] \cdots \mathbb{E} \left[\frac{S_{K-1}}{S_{K-2}} \right] \cdot \mathbb{E} \left[\frac{S_K}{S_{K-1}} \right] \\ &= \left(1 + \frac{\mu_D}{K} \right)^K, \end{aligned} \quad (8.6.8)$$

where we have used independence of $\frac{S_1}{S_0}, \frac{S_2}{S_1}, \dots, \frac{S_{K-1}}{S_{K-2}}, \frac{S_K}{S_{K-1}}$. Lemma 8.4.1 implies that

$$\lim_{K \rightarrow \infty} \mathbb{E} \left[\frac{S_K}{S_0} \right] = e^{\mu_D}. \quad (8.6.9)$$

If we take the logarithm before taking the limit, we can use the Taylor series expansion (8.4.27) of logarithm to write

$$\begin{aligned}
\ln \mathbb{E} \left[\frac{S_K}{S_0} \right] &= \ln \left(1 + \frac{\mu_D}{K} \right)^K \\
&= K \ln \left(1 + \frac{\mu_D}{K} \right) \\
&= K \left(\frac{\mu_D}{K} + O \left(\frac{1}{K^2} \right) \right) \\
&= \mu_D + O \left(\frac{1}{K} \right).
\end{aligned} \tag{8.6.10}$$

We conclude that

$$\lim_{K \rightarrow \infty} \ln \mathbb{E} \left[\frac{S_K}{S_0} \right] = \mu_D. \tag{8.6.11}$$

Equation (8.6.11) can also be derived directly from (8.6.9) by using the fact that natural logarithm is a continuous function.

With p and q given by (8.6.6), we now compute the expected rate of growth, the right-hand side of (8.6.2). For $k = 1, 2, \dots, K$, we have

$$\mathbb{E} \left[\ln \frac{S_k}{S_{k-1}} \right] = p \ln u + q \ln d.$$

Therefore,

$$\begin{aligned}
\mathbb{E} \left[\ln \frac{S_K}{S_0} \right] &= \mathbb{E} \left[\ln \left(\frac{S_1}{S_0} \cdot \frac{S_2}{S_1} \cdots \frac{S_{K-1}}{S_{K-2}} \cdot \frac{S_K}{S_{K-1}} \right) \right] \\
&= \mathbb{E} \left[\ln \frac{S_1}{S_0} + \ln \frac{S_2}{S_1} + \cdots + \ln \frac{S_{K-1}}{S_{K-2}} + \ln \frac{S_K}{S_{K-1}} \right] \\
&= \mathbb{E} \left[\ln \frac{S_1}{S_0} \right] + \mathbb{E} \left[\ln \frac{S_2}{S_1} \right] + \cdots + \mathbb{E} \left[\ln \frac{S_{K-1}}{S_{K-2}} \right] + \mathbb{E} \left[\ln \frac{S_K}{S_{K-1}} \right] \\
&= K(p \ln u + q \ln d).
\end{aligned} \tag{8.6.12}$$

We again use the Taylor series expansion (8.4.27) of $\ln(1+x)$ to continue

the computation. From this expansion, (8.6.12), and (8.6.7), we have

$$\begin{aligned}
\mathbb{E} \left[\ln \frac{S_K}{S_0} \right] &= K \left[p \ln \left(1 + \frac{r_D}{K} + \frac{\sigma_D}{\sqrt{K}} \right) + q \ln \left(1 + \frac{r_D}{K} - \frac{\sigma_D}{\sqrt{K}} \right) \right] \\
&= K \left[p \left(\frac{r_D}{K} + \frac{\sigma_D}{\sqrt{K}} - \frac{\sigma_D^2}{2K} + O \left(\frac{1}{K\sqrt{K}} \right) \right) \right. \\
&\quad \left. + q \left(\frac{r_D}{K} - \frac{\sigma_D}{\sqrt{K}} - \frac{\sigma_D^2}{2K} + O \left(\frac{1}{K\sqrt{K}} \right) \right) \right] \\
&= K \left[\frac{r_D - \frac{1}{2}\sigma_D^2}{K} + \frac{(p-q)\sigma_D}{\sqrt{K}} \right] + O \left(\frac{1}{\sqrt{K}} \right) \\
&= r_D - \frac{1}{2}\sigma_D^2 + \mu_D - r_D + O \left(\frac{1}{\sqrt{K}} \right) \\
&= \mu_D - \frac{1}{2}\sigma_D^2 + O \left(\frac{1}{\sqrt{K}} \right). \tag{8.6.13}
\end{aligned}$$

Therefore,

$$\lim_{K \rightarrow \infty} \mathbb{E} \left[\ln \frac{S_K}{S_0} \right] = \mu_D - \frac{1}{2}\sigma_D^2. \tag{8.6.14}$$

Note from (8.6.11) and (8.6.14) that the Jensen inequality prediction (8.6.3) does indeed hold, and the size of the gap depends on the volatility parameter σ_D — a small σ_D results in a small gap and a large σ_D results in a large gap.

Suppose we buy one share of stock at time zero and hold it for L days, where L is some large number. Which formula (8.6.11) or (8.6.14) is the better representation of what we should expect of this investment? We answer this question here.

We are tossing the coin K times per day, and in L days we would toss the coin KL times. We define $N = KL$. At the beginning of day zero we invest S_0 , and at the end of day L we have stock valued at S_N , which is a random number (depending on the coin tossing). We define the random number ν_D by the equation

$$S_N = S_0 e^{\nu_D L}. \tag{8.6.15}$$

If instead of buying one share of stock at time zero, we had instead invested the money at a continuously compounding daily interest rate ν_D , then we would have the same amount of money after L days as we have obtained

from the stock investment. Solving (8.5.16) for ν_D , we obtain

$$\begin{aligned}\nu_D &= \frac{1}{L} \ln \frac{S_N}{S_0} \\ &= \frac{1}{L} \left[\ln \frac{S_K}{S_0} + \ln \frac{S_{2K}}{S_K} + \ln \frac{S_{3K}}{S_{2K}} + \cdots + \ln \frac{S_{LK}}{S_{(L-1)K}} \right].\end{aligned}\quad (8.6.16)$$

The random variables $\ln \frac{S_K}{S_0}, \ln \frac{S_{2K}}{S_K}, \dots, \ln \frac{S_{LK}}{S_{(L-1)K}}$ are the rates of return on days one, two, etc., up to day K . These are independent of one another (they depend on coin tosses on different days), they each have the same distribution, and, in particular, they all have the same expected value. According to (8.6.14), the expected value of each of these random variables is approximately $\mu_D - \frac{1}{2}\sigma_D^2$. The Law of Large Numbers implies that for large values of L , the right-hand side of (8.6.16) is approximately also equal to $\mu_D - \frac{1}{2}\sigma_D^2$. Therefore,

$$\nu_D \approx \mu_D - \frac{1}{2}\sigma_D^2, \quad (8.6.17)$$

and formula (8.6.14) gives the more accurate description of what we can expect of our investment in the stock. If we hold the stock over several days (40 or more), it will achieve a daily average growth rate approximately equal to $\mu_D - \frac{1}{2}\sigma_D^2$. The stock volatility reduces the rate of growth from what we might expect from equation (8.6.5) or (8.6.11).

8.6.2 Estimating volatility

In order to build the binomial model discussed in this section, we need to know the parameter σ_D . We have just observed that the average growth rate of the stock over several days is an estimate $\mu_D - \frac{1}{2}\sigma_D^2$, but since we do not know μ_D , this does not help us determine σ_D . In this section, we present a procedure for estimating σ_D that does not require us to estimate μ_D .

Recall the random variables X_k of (8.2.2):

$$X_k = \begin{cases} +1 & \text{if the } k\text{-th coin toss results in } H, \\ -1 & \text{if the } k\text{-th coin toss results in } T. \end{cases} \quad (8.2.2)$$

Let H_K be the number of heads obtained in the first K coin tosses and let T_K be the number of tails. Define

$$M_K = H_K - T_K = \sum_{k=1}^K X_k. \quad (8.6.18)$$

If we use the risk-neutral probabilities $\tilde{p} = \tilde{q} = \frac{1}{2}$, the Central Limit Theorem tells us that the distribution of $\frac{M_K}{\sqrt{K}}$ approaches the normal distribution as $K \rightarrow \infty$ (see Table 8.3.1). However, in this section we have probabilities p and q for head and tail given by (8.6.6). Under these probabilities, instead of X_k having expected value zero,

$$\mathbb{E}X_k = p - q = \frac{\mu_D - r_D}{\sigma_D \sqrt{K}}. \quad (8.6.19)$$

Despite the difference created by the choice of probabilities, we can still follow the development of (8.4.28) to obtain a formula for $\ln S_K$. In particular,

$$\begin{aligned} \ln S_K &= \ln(S_0 u^{H_K} d^{T_K}) \\ &= \ln S_0 + H_K \ln u + T_K \ln d \\ &= \ln S_0 + H_K \left(\frac{r_D}{K} + \frac{\sigma_D}{\sqrt{K}} - \frac{\sigma_D^2}{2K} + O\left(\frac{1}{K\sqrt{K}}\right) \right) \\ &\quad + T_K \left(\frac{r_D}{K} - \frac{\sigma_D}{\sqrt{K}} - \frac{\sigma_D^2}{2K} + O\left(\frac{1}{K\sqrt{K}}\right) \right) \\ &= \ln S_0 + r_D - \frac{1}{2}\sigma_D^2 + \sigma_D \frac{M_K}{\sqrt{K}} + O\left(\frac{1}{\sqrt{K}}\right). \end{aligned} \quad (8.6.20)$$

Here we have used the fact that $H_K + T_K = K$ and also used the facts that $0 \leq H_K \leq K$ and $0 \leq T_K \leq K$ to justify the relations

$$H_K O\left(\frac{1}{K\sqrt{K}}\right) = O\left(\frac{1}{\sqrt{K}}\right), \quad T_K O\left(\frac{1}{K\sqrt{K}}\right) = O\left(\frac{1}{\sqrt{K}}\right).$$

We define

$$Y_1 = \frac{M_K}{\sqrt{K}} = \frac{1}{\sqrt{K}} \sum_{k=1}^K X_k. \quad (8.6.21)$$

According to (8.6.19),

$$\mathbb{E}Y_1 = \frac{\mu_D - r_D}{\sigma_D}. \quad (8.6.22)$$

To determine the variance of Y_1 , we recall that the variance of a sum of

independent random variables is the sum of the variances, and hence

$$\begin{aligned}
\text{Var}(M_K) &= \sum_{k=1}^K \text{Var}(X_k) \\
&= \sum_{k=1}^K \left(\mathbb{E}[X_k^2] - (\mathbb{E}[X_k])^2 \right) \\
&= \sum_{k=1}^K \left(1 - \frac{(\mu_D - r_D)^2}{\sigma_D^2 K} \right) \\
&= K - \frac{(\mu_D - r_D)^2}{\sigma_D^2}.
\end{aligned}$$

Therefore,

$$\text{Var}(Y_1) = \frac{1}{K} \text{Var}(M_K) = 1 - \frac{(\mu_D - r_D)^2}{K\sigma_D^2}. \quad (8.6.23)$$

Using Y_1 , we may rewrite (8.6.20) as

$$\ln \frac{S_K}{S_0} = r_D - \frac{1}{2}\sigma_D^2 + \sigma_D Y_1 + O\left(\frac{1}{\sqrt{K}}\right). \quad (8.6.24)$$

We can observe the left-hand side of this equation from market data, and we wish to estimate the parameter σ_D appearing on the right-hand side. To estimate σ_D , we consider L trading days and write an equation like (8.6.24) for each trading day. The stock price at the end of the $(\ell - 1)$ -st trading day is $S_{(\ell-1)K}$ and the stock price at the end of the ℓ -th trading day is $S_{\ell K}$. We observe these prices in the market and then compute the natural logarithm of the ratio, which we call $Z_\ell = \ln \frac{S_{\ell K}}{S_{(\ell-1)K}}$. Analogous to (8.6.24), we have

$$Z_\ell = \ln \frac{S_{\ell K}}{S_{(\ell-1)K}} = r_D - \frac{1}{2}\sigma_D^2 + \sigma_D Y_\ell + O\left(\frac{1}{\sqrt{K}}\right), \quad \ell = 1, 2, \dots, L. \quad (8.6.25)$$

where

$$Y_\ell = \frac{1}{\sqrt{K}} \sum_{k=(\ell-1)K+1}^{\ell K} X_k \quad (8.6.26)$$

has expected value and variance

$$\mathbb{E}Y_\ell = \frac{\mu_D - r_D}{\sigma_D}, \quad \text{Var}(Y_\ell) = 1 - \frac{(\mu_D - r_D)^2}{K\sigma_D^2}. \quad (8.6.27)$$

Furthermore, $Y_1, Y_2, \dots, Y_L$ are independent because they depend on coin tosses on different days.

The variance of each Z_ℓ is $\sigma_D^2 \text{Var}(Y_\ell) = \sigma_D^2 - \frac{(\mu_D - r_D)^2}{K}$, and so the standard deviation, the square root of the variance, is $\sigma_D + O(\frac{1}{K})$ (see Exercise 8.6). The usual estimator of the common standard deviation of L independent and indentially distributed random variables $Z_1, Z_2, \dots, Z_L$ is

$$\hat{\sigma}_D = \sqrt{\frac{1}{L-1} \sum_{\ell=1}^L Z_\ell^2 - \frac{1}{L(L-1)} \left(\sum_{\ell=1}^L Z_\ell \right)^2}. \quad (8.6.28)$$

This formula provides an estimate of $\sigma_D + O(\frac{1}{K})$ based on market data. Since K is large, we can use the expression in (8.6.28) as an estimate of σ_D .

8.6.3 Daily parameters and annual parameters

In this section, we have divided one day into K periods and used the up and down factors (8.6.1) in each period, where r_D and σ_D are the daily interest rate and daily volatility, respectively. Usually the interest rate for the money market account and the volatility of a stock are quoted on an annual rather than a daily basis. Let L be the number of days in a year. Then in a year's time there are $N = KL$ periods. If r and σ are the annual interest rate and annual volatility, we would use as up and down factors (see (8.4.16) with τ equal one year)

$$u = 1 + \frac{r}{N} + \frac{\sigma}{\sqrt{N}} = 1 + \frac{r/L}{K} + \frac{\sigma/\sqrt{L}}{\sqrt{K}}, \quad (8.6.29)$$

$$d = 1 + \frac{r}{N} - \frac{\sigma}{\sqrt{N}} = 1 + \frac{r/L}{K} - \frac{\sigma/\sqrt{L}}{\sqrt{K}}. \quad (8.6.30)$$

Comparing (8.6.29)–(8.6.30) to (8.6.1), we see that the relationship between the daily interest rate r_D and the annual interest rate r is

$$r_D = \frac{r}{L}, \quad (8.6.31)$$

and the relationship between the daily volatility σ_D and the annual volatility σ is

$$\sigma_D = \frac{\sigma}{\sqrt{L}}. \quad (8.6.32)$$

The number of days in a non-leap year is 365, but experience has shown that using $L = 365$ in equation (8.6.32) is not a good idea. Stock prices can

move only when the exchanges on which they are traded are open. Because exchanges are closed on week-ends and holidays, the number of trading days in a year is 252, give or take a day or two depending on the year. Thus $L = 252$ is a good number to use in (8.6.32). One could use $L = 365$ in (8.6.31) since interest accumulates on week-ends as well as weekdays, but when one then builds a model with 252 trading day periods in a year, the amount of interest accumulating in one year of the model would be only $(1 + \frac{r}{365})^{252}$ rather than the desired $(1 + \frac{r}{365})^{365}$. The simplest way around this dilemma is to take $L = 252$ in (8.6.31) as well, and pretend that interest accumulates slightly faster than $\frac{r}{365}$ on trading days and does not accumulate at all on non-trading days. The error introduced by this simplification is small and is usually ignored. With $r = 0.05$,

$$\begin{aligned} \left(1 + \frac{r}{365}\right)^{365} &= 1.0512675, \\ \left(1 + \frac{r}{252}\right)^{252} &= 1.0512659, \end{aligned}$$

whereas

$$\left(1 + \frac{r}{365}\right)^{252} = 1.0351209.$$

In summary, the usual procedure to estimate a stock volatility from historical data is to observe the closing stock price on L successive trading days, then compute Z_ℓ , the natural logarithm of the the closing price on day ℓ divided by the closing price on day $\ell - 1$, and then use these L observations $Z_1, Z_2, \dots, Z_L$ in (8.6.28) to compute $\hat{\sigma}_D$. This is an estimate of the daily volatility σ_D . One then estimates the annual volatility to be

$$252 \times \hat{\sigma}_D.$$

The number of observations is not required to be 252.

8.6.4 Implied volatility

The estimation of annual volatility described in Subsection 8.6.3 leads to a number that can be used in the Black-Scholes formula to compute prices of put and call options.

If options on a particular stock are not actively traded, this is a good procedure to find a theoretical price. If options on a stock are actively traded, then one can use the option prices themselves to “imply” a volatility.

Typically there are options with different strike prices K and different times to expiration τ , and for each such option one can determine the volatility $\sigma(\tau, K)$ that makes the Black-Scholes formula agree with the market price of the option. If the assumptions of the Black-Scholes model were completely correct, the implied volatilities $\sigma(\tau, K)$ would not depend on τ and K . In fact, they often do and the dependence is sometimes quite pronounced.

Even in a case where the dependence of implied volatilities on time to expiration and/or strike prices is pronounced, the Black-Scholes formula provides a useful interpolation device. Suppose there are market prices on options with time to expiration τ and strike price K_0 and also on options with the same time to expiration but with a different strike price K_2 , where $K_2 > K_0$. Suppose K_1 is between K_0 and K_2 , options with strike price K_1 are not being traded, and you are asked to quote a price on an option with time to expiration τ and strike price K_1 . You can imply the volatilities $\sigma(\tau, K_0)$ and $\sigma(\tau, K_2)$ from the market prices, take a weighted average of these as the volatility for the option with strike price K_1 , and compute the price of this option using the Black-Scholes formula. In addition to using the Black-Scholes formula to compute the price of the option, you can also use it to determine how to hedge the option by the procedure described in Subsection 8.5.5.